

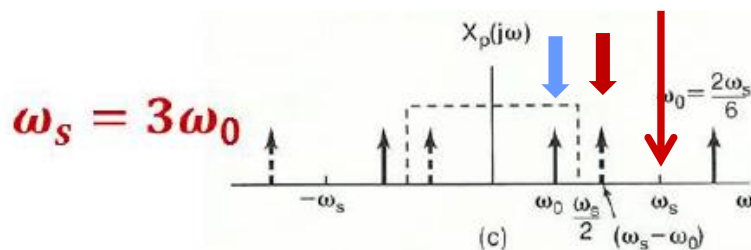
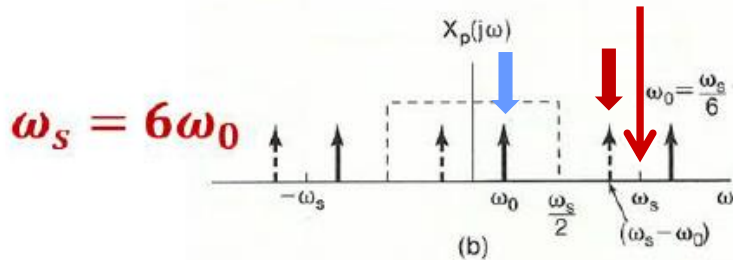
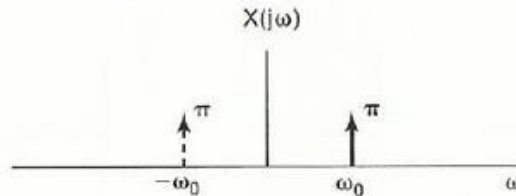
- Homework: 8.4, 8.25, 8.29
- Tutorial: 8.34, 8.35, 8.38

Aliasing: Example

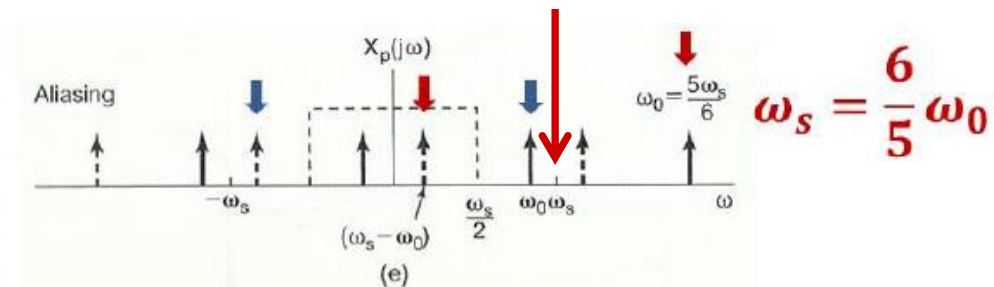
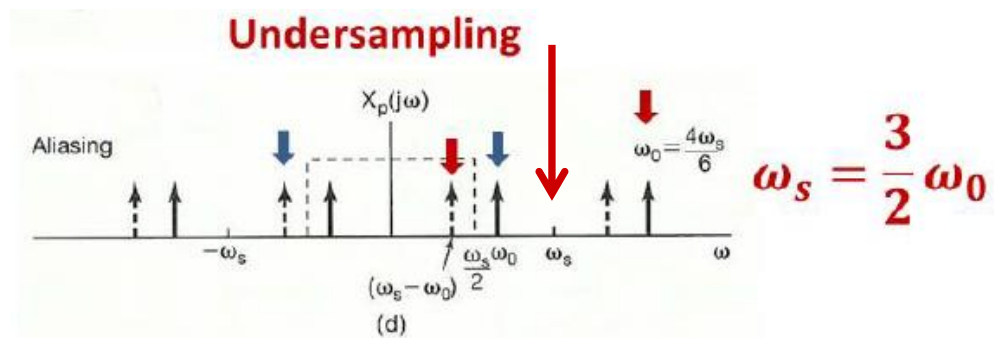
Signal before sampling: $\cos \omega_0 t$

Sampling rate: ω_s

Lowpass Filter: $\frac{-\omega_s}{2} \sim \frac{\omega_s}{2}$

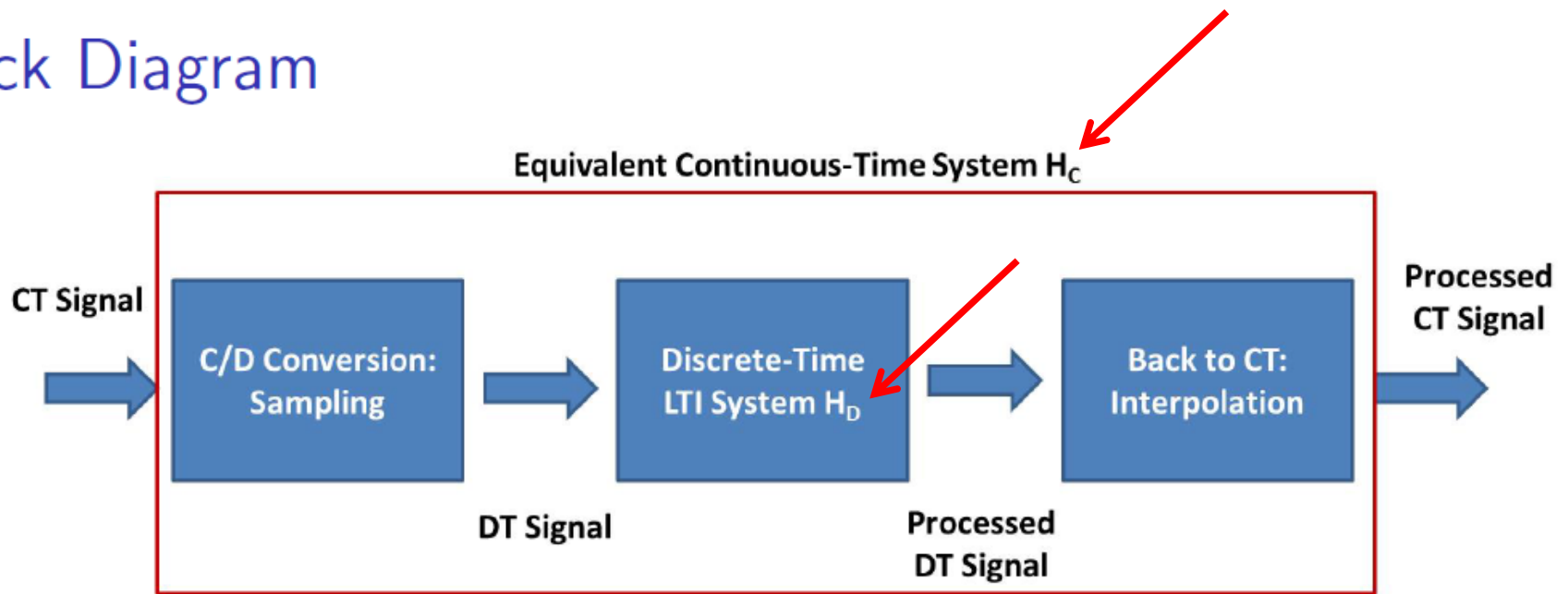


$\cos \omega_0 t$



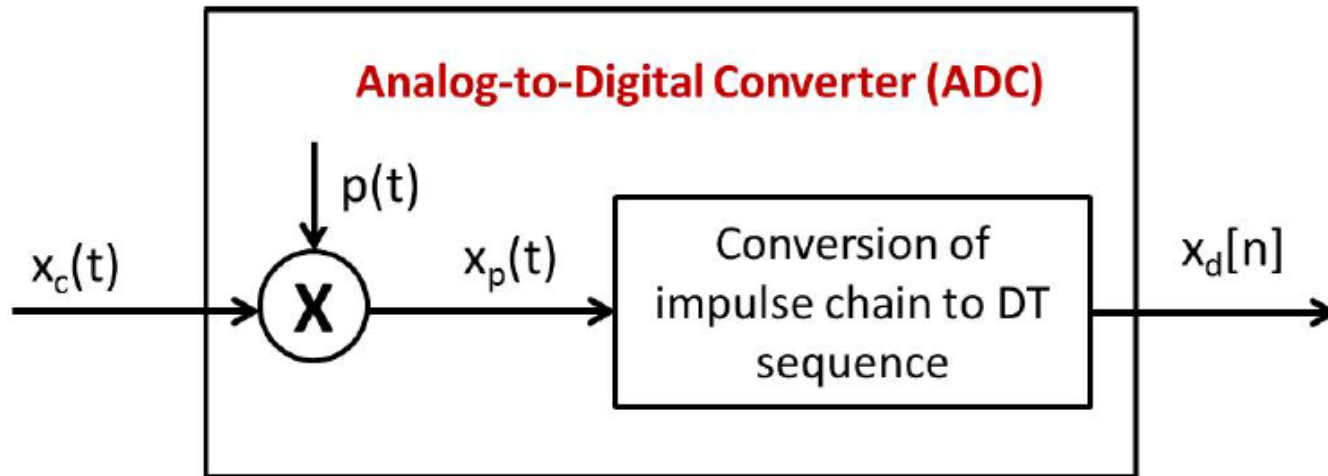
Aliasing: $\cos(\omega_s - \omega_0)t$

Block Diagram



- It is much easier to design DT system.
- What's the relation between H_C and H_D ?

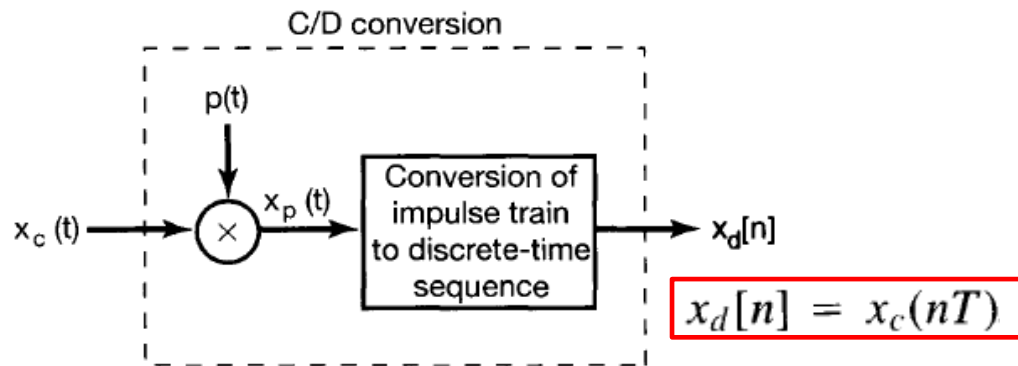
Discretization: C/D Conversion



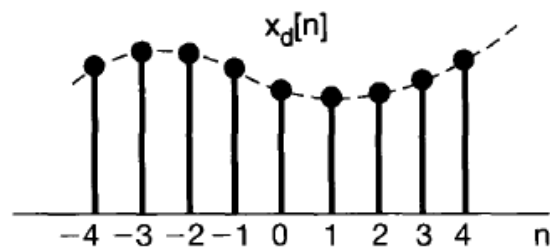
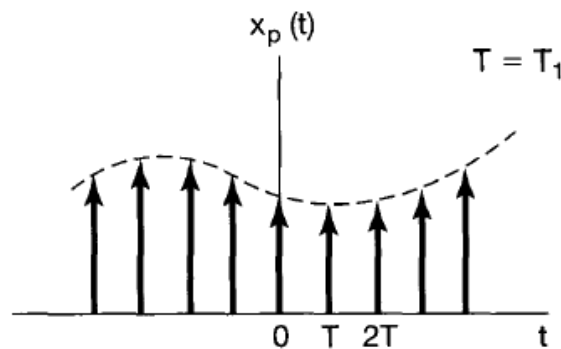
- Mathematical Interpretation (Fourier Transform)

$$x_c(t) \longleftrightarrow X_c(j\omega)$$

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x_c(nT)\delta(t - nT) \longleftrightarrow X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$



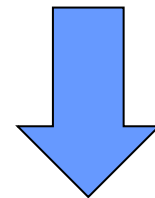
(a)



$$X_d(e^{j\Omega}) = \sum_{n=-\infty}^{+\infty} x_d[n] e^{-j\Omega n}$$

$$= \sum_{n=-\infty}^{+\infty} x_c(nT) e^{-j\Omega n}$$

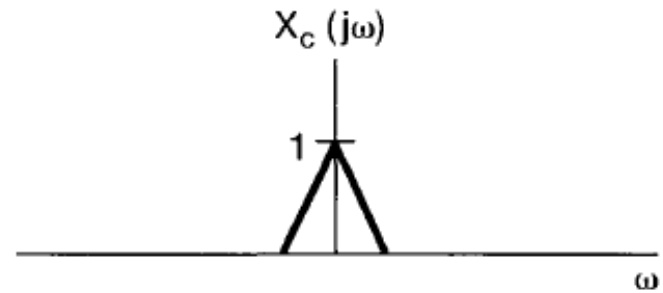
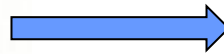
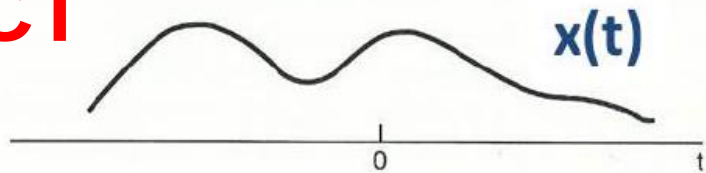
$$X_p(j\omega) = \sum_{n=-\infty}^{+\infty} x_c(nT) e^{-j\omega nT}$$



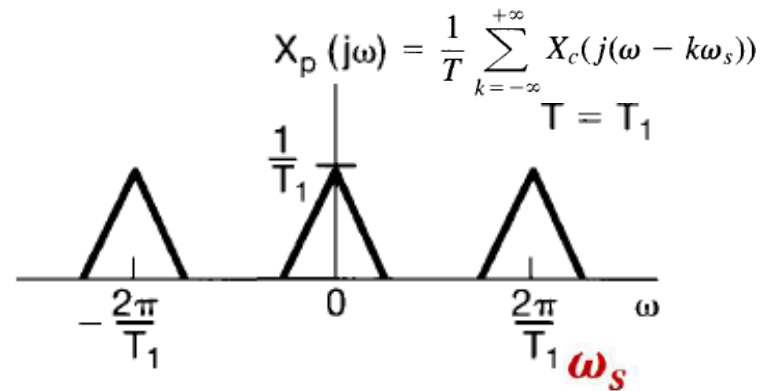
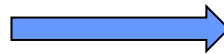
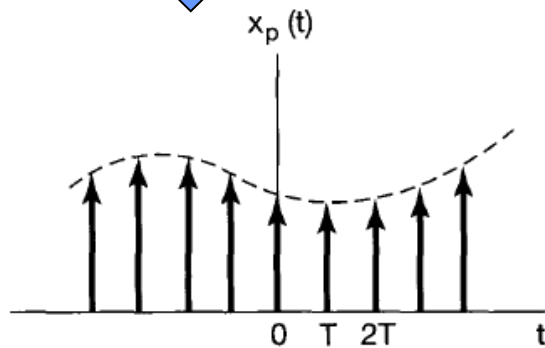
$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$



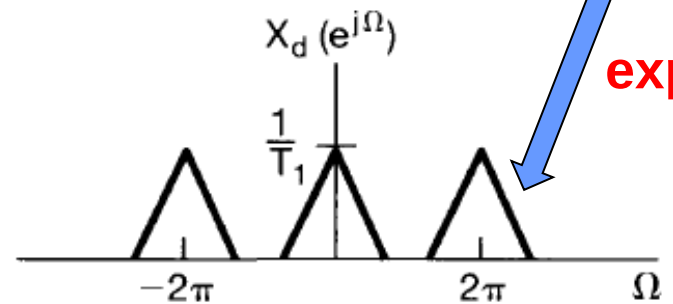
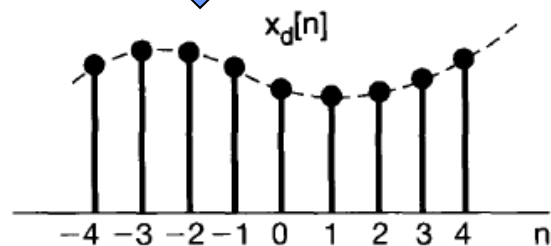
CT



CT



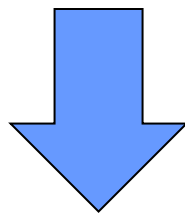
DT



$$X_d(e^{j\Omega}) = X_p(j\Omega/T)$$

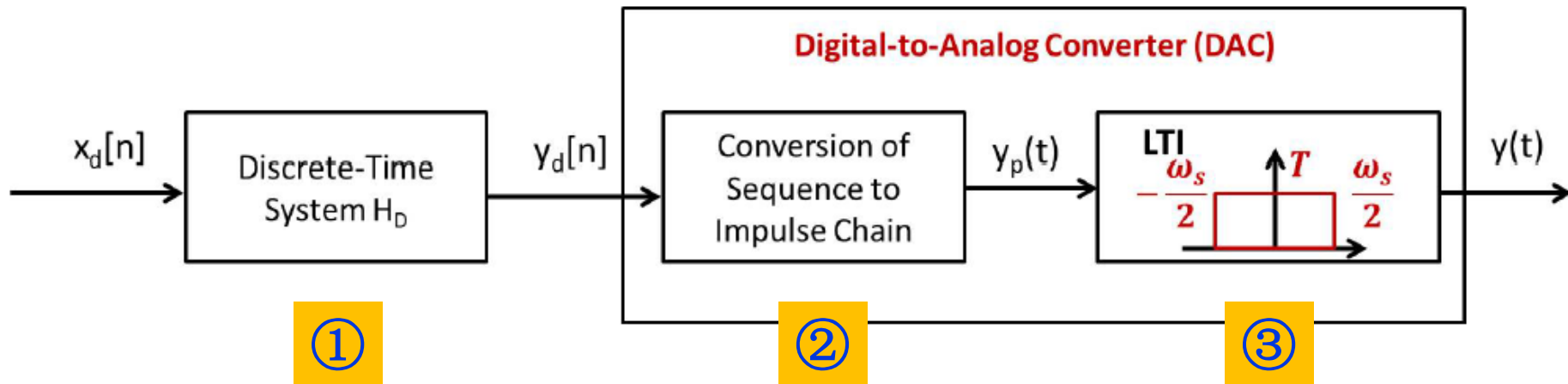
$$X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s))$$

$$X_d(e^{j\omega}) = X_p(j\omega/T)$$



$$\begin{aligned} X_d(e^{j\omega}) &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s)) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega - 2\pi k}{T})) \end{aligned}$$

DT Processing and Conversion



• Mathematical Interpretation (Fourier Transform)

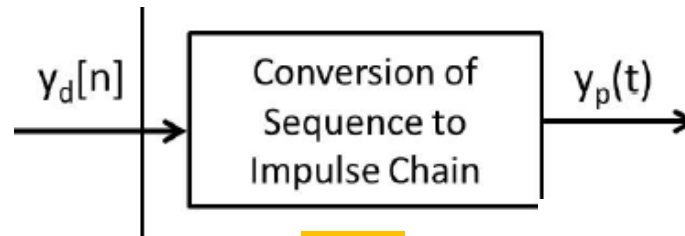
$$\textcircled{1} \quad y_d[n] = x_d[n] * h_D[n] \quad \longleftrightarrow \quad Y_d(e^{j\omega}) = X_d(e^{j\omega})H_D(e^{j\omega})$$

$$\textcircled{2} \quad y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n]\delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$

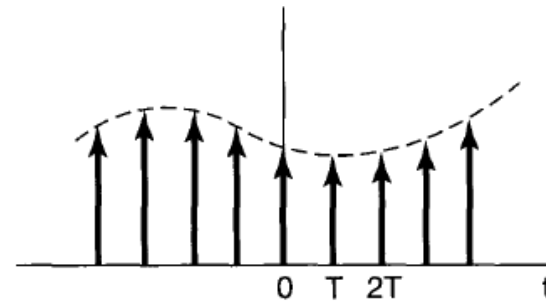
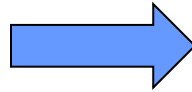
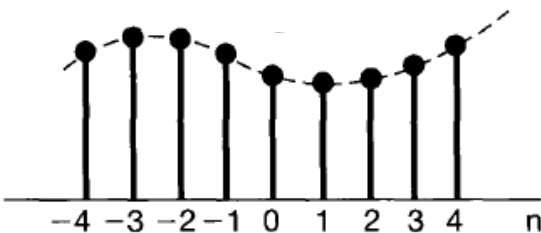
$$\textcircled{3} \quad y(t) = y_p(t) * h_{LP}(t) \quad \longleftrightarrow \quad Y(j\omega) = Y_p(j\omega)H_{LP}(j\omega)$$

$$Y(j\omega) = X_d(e^{j\omega T})H_D(e^{j\omega T})H_{LP}(j\omega)$$

$$y_d[n] = y_c(nT)$$

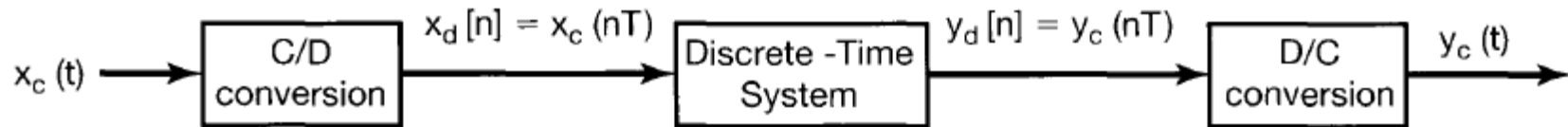


②



1. $n \rightarrow t$
2. time-scaling

$$y_p(t) = \sum_{n=-\infty}^{\infty} y_d[n] \delta(t - nT) \quad \longleftrightarrow \quad Y_p(j\omega) = Y_d(e^{j\omega T})$$



$$Y(j\omega) = X_d(e^{j\omega T}) H_D(e^{j\omega T}) H_{LP}(j\omega)$$

$$X_d(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\frac{\omega}{T} - k\omega_s))$$

$$Y(j\omega) = \left[\frac{1}{T} \sum_{k=-\infty}^{+\infty} X_c(j(\omega - k\omega_s)) \right] H_D(e^{j\omega T}) H_{LP}(j\omega)$$

$$= X_c(j\omega) H_D(e^{j\omega T})$$

$$= X_c(j\omega) \tilde{H}_D(e^{j\omega T}) \quad (1)$$

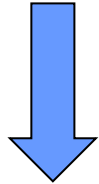
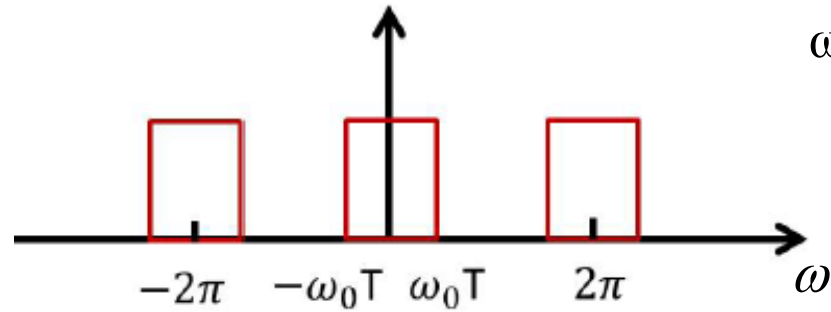
where

$$\tilde{H}_D(e^{j\omega T}) = \begin{cases} H_D(e^{j\omega T}) & |\omega| < \omega_s/2 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

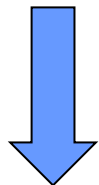
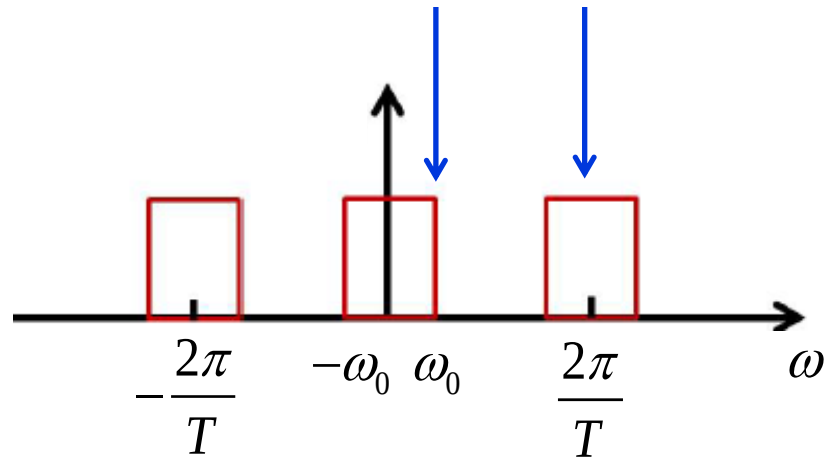
- It is equivalent to a continuous-time LTI system $H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$
- $H_D(e^{j\omega T})$ is a periodic extension of $\tilde{H}_D(e^{j\omega T})$ with period $\omega_s = 2\pi/T$

T : sampling period
 $\omega_s = 2\pi/T$: sampling rate

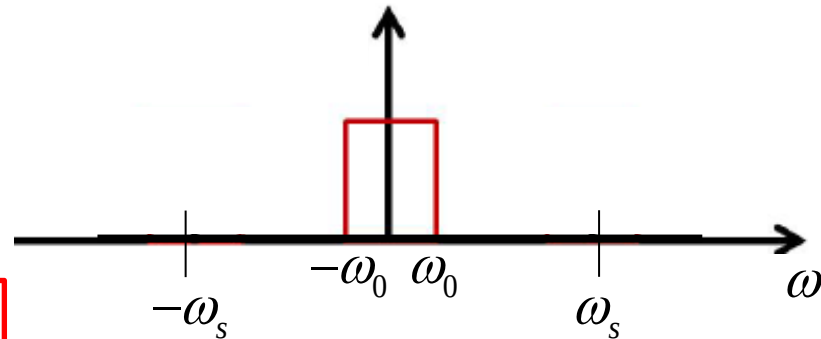
$$H_D(e^{j\omega})$$



$$H_D(e^{j\omega T})$$



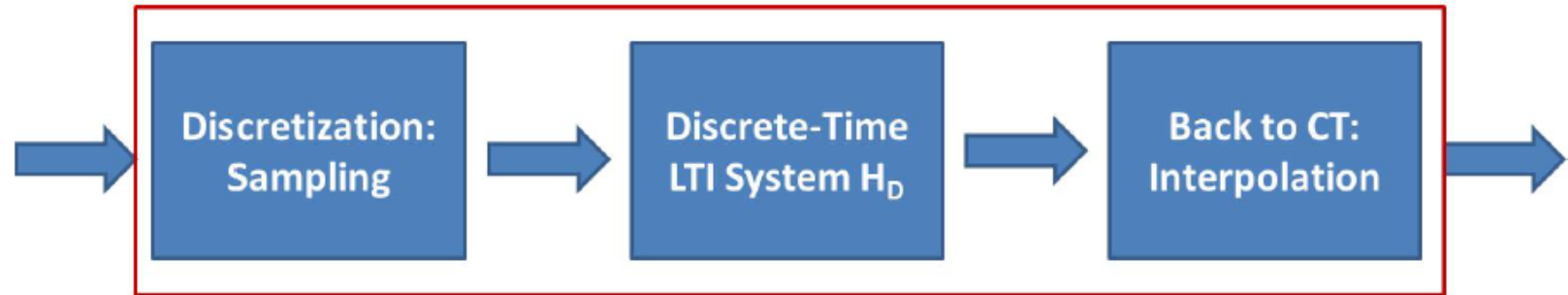
$$\tilde{H}_D(e^{j\omega T})$$



$$H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$$

System Design

$$H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$$

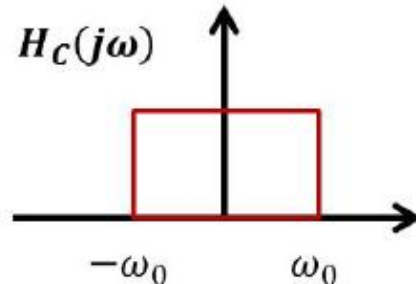


- How can we design a CT LTI system with frequency response H_C via DT LTI system?
- Step 1: Sampling frequency ω_s or $2\pi/T$ should be larger than Nyquist rate
- Step 2: $\tilde{H}_D(e^{j\omega T}) = H_C(j\omega)$
- Step 3: Frequency response of DT LTI system
 $H_D(e^{j\omega T}) = \sum_{k=-\infty}^{\infty} \tilde{H}_D(e^{j(\omega - k\omega_s)T})$ or
 $H_D(e^{j\omega}) = \sum_{k=-\infty}^{\infty} \tilde{H}_D(e^{j(\omega - k\omega_s T)}) = \sum_{k=-\infty}^{\infty} H_C(j\frac{\omega - 2k\pi}{T})$

System Design Example

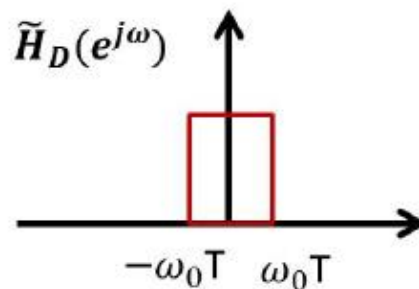
- How to implement an ideal CT lowpass filter?

Objective of Design:

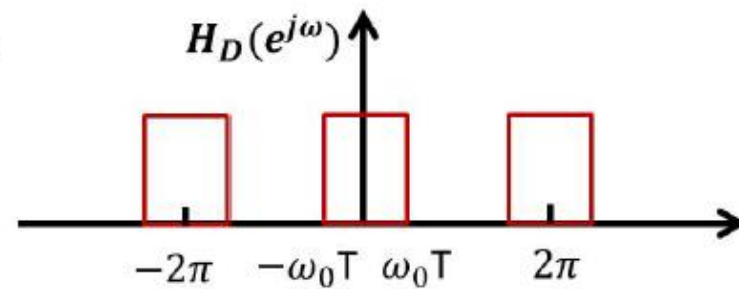


$$H_C(j\omega) = \tilde{H}_D(e^{j\omega T})$$

Scale by T



Repetition

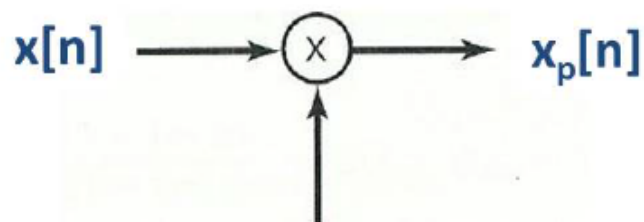


Practice

- Design a DT system to process a CT signal with ideal low-pass filtering
 - ◆ cutoff frequency: $f_{\text{cut}}=50$ Hz
 - ◆ Sampling frequency: $f_s=1000$ Hz

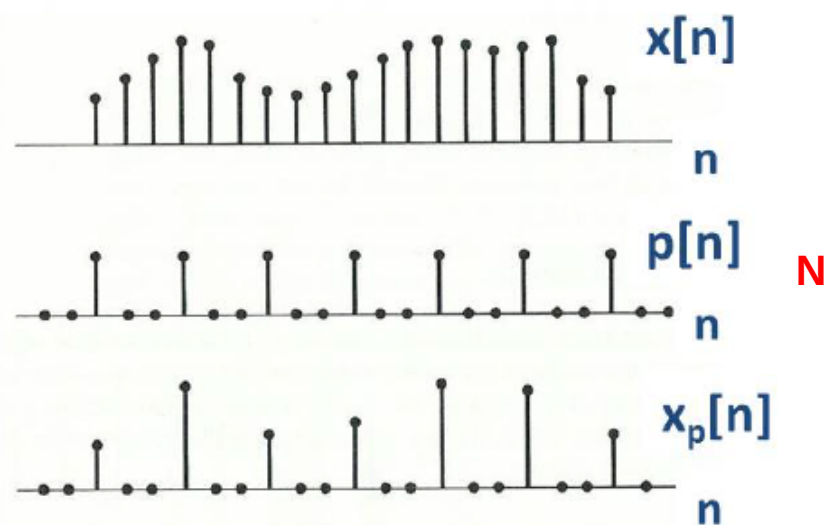
Sampling on Discrete-Time Signals

- In digital signal processing, we may sample the discrete-time signals
- Example: image, audio, video compression
- Its mathematical model is give below:



$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$



Frequency Analysis

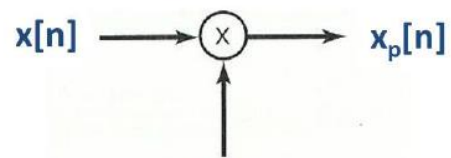
$$P(e^{j\omega}) = \frac{2\pi}{N} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s) \text{ where } \omega_s = 2\pi/N$$

$$X_p(e^{j\omega}) = \frac{1}{2\pi} \int_{2\pi} P(e^{j\theta}) X(e^{j(\omega-\theta)}) d\theta = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega-k\omega_s)})$$

periodic convolution

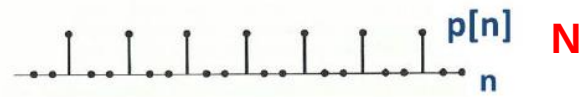
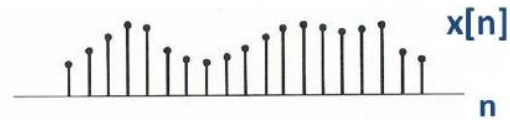
$$= \frac{1}{N} [X(e^{j\omega}) + X(e^{j(\omega-\omega_s)}) + \dots + X(e^{j(\omega-(N-1)\omega_s})]$$

What is the relation between $X(e^{j\omega})$ and $X_p(e^{j\omega})$?

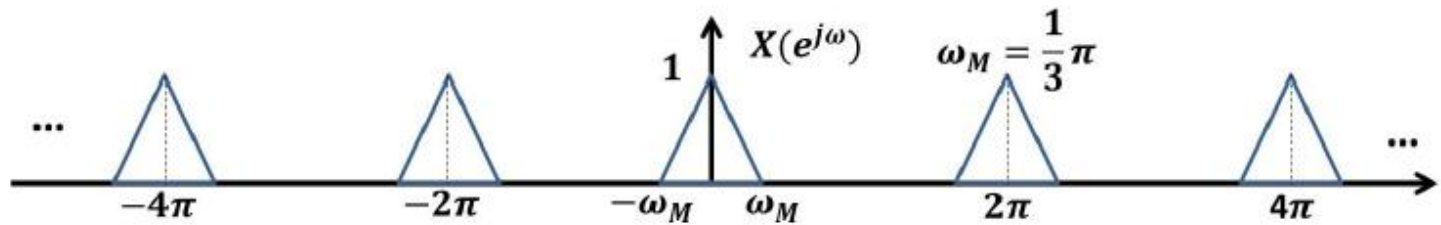


$$p[n] = \sum_{k=-\infty}^{+\infty} \delta[n - kN]$$

$$x_p[n] = \sum_{k=-\infty}^{+\infty} x[kN] \delta[n - kN]$$

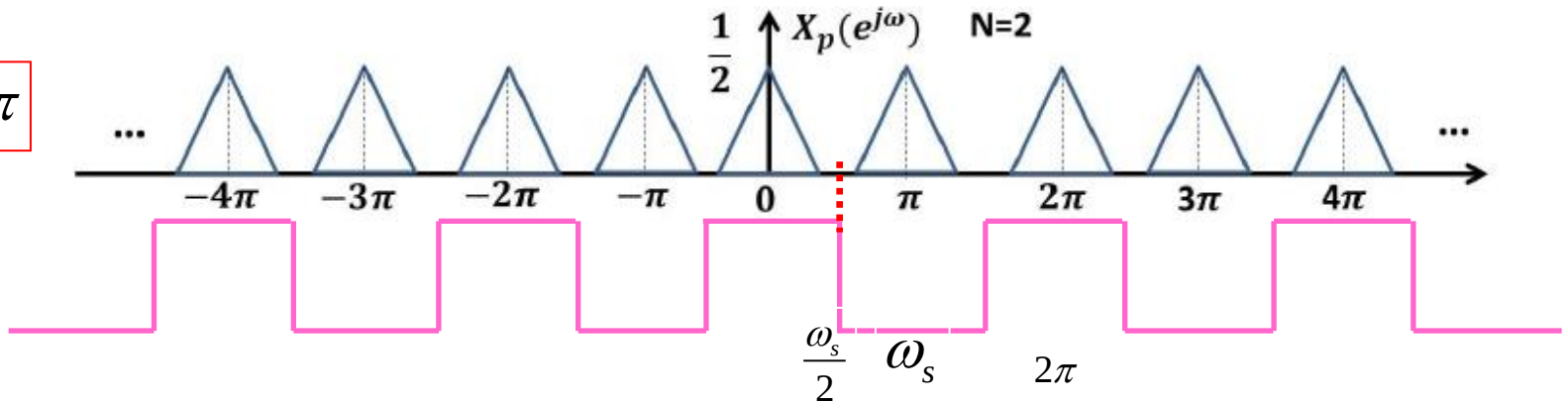


$$\omega_s = 2\pi$$



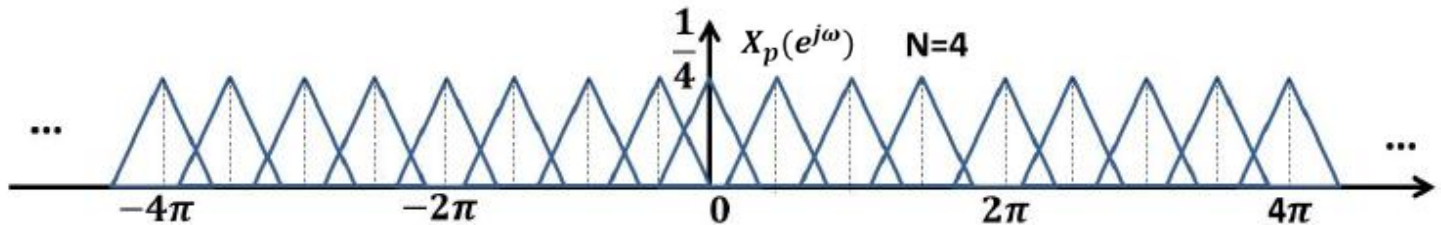
$$N=2$$

$$\omega_s = \pi$$



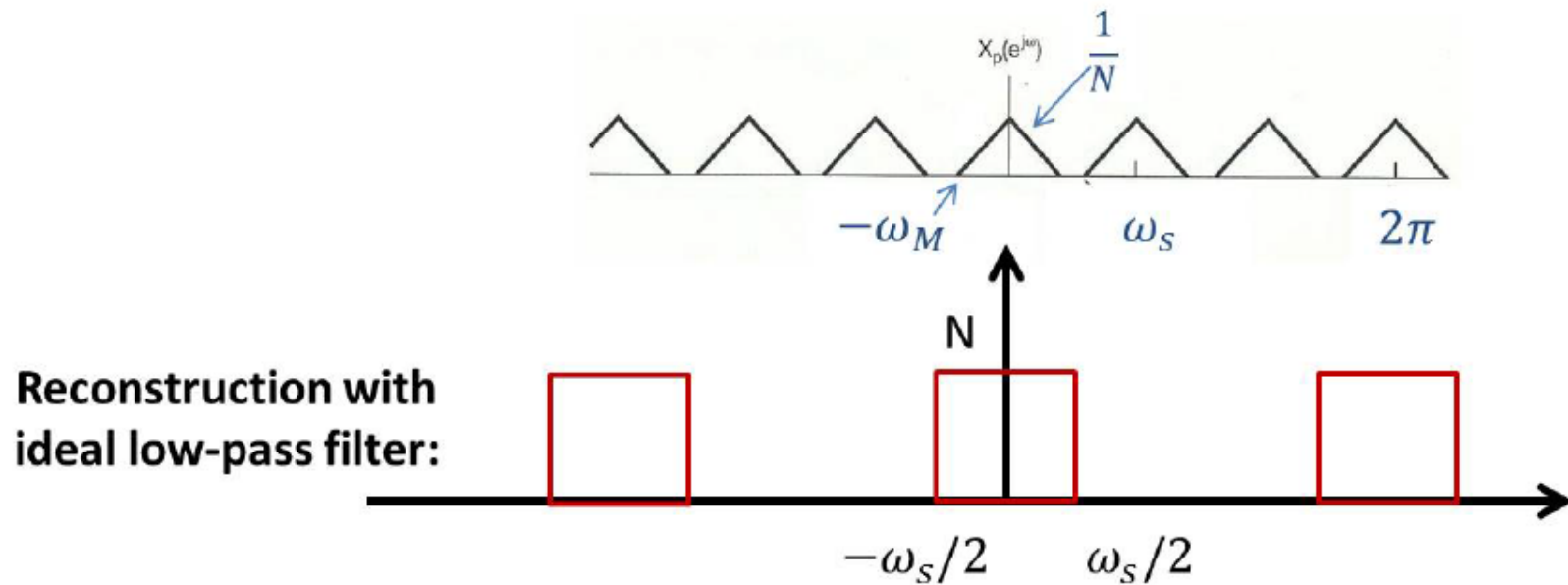
$$N=4$$

$$\omega_s = \frac{\pi}{2}$$



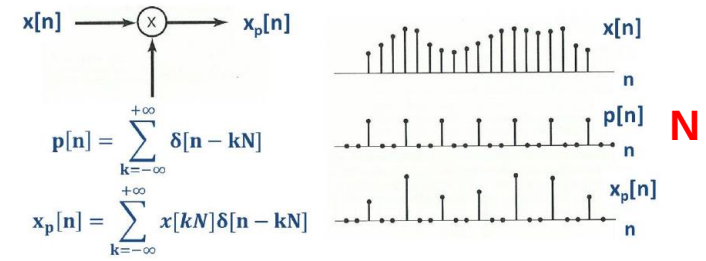
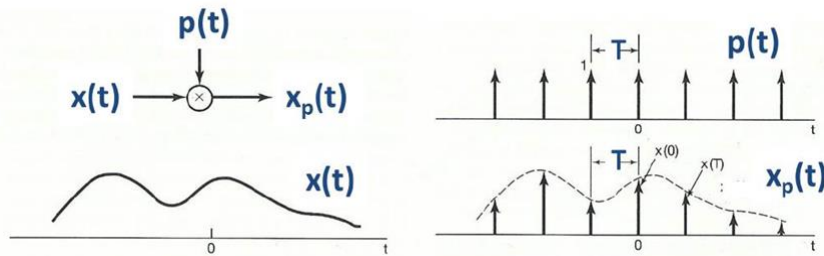
Reconstruction

- Perfect reconstruction is applicable when $\omega_s > 2\omega_M \leftrightarrow N < \frac{\pi}{\omega_M}$



- Aliasing occurs when $\omega_s < 2\omega_M$

Comparison btw CT sampling and DT sampling



Both have sampling theory

CT

Sampling period/frequency

$$T, \omega_s = 2\pi/T$$

Sampling theory

$$\omega_s > 2\omega_M$$

Reconstruction with ideal LPF

$$[-\omega_s/2 \quad \omega_s/2], \text{ gain } T$$

DT

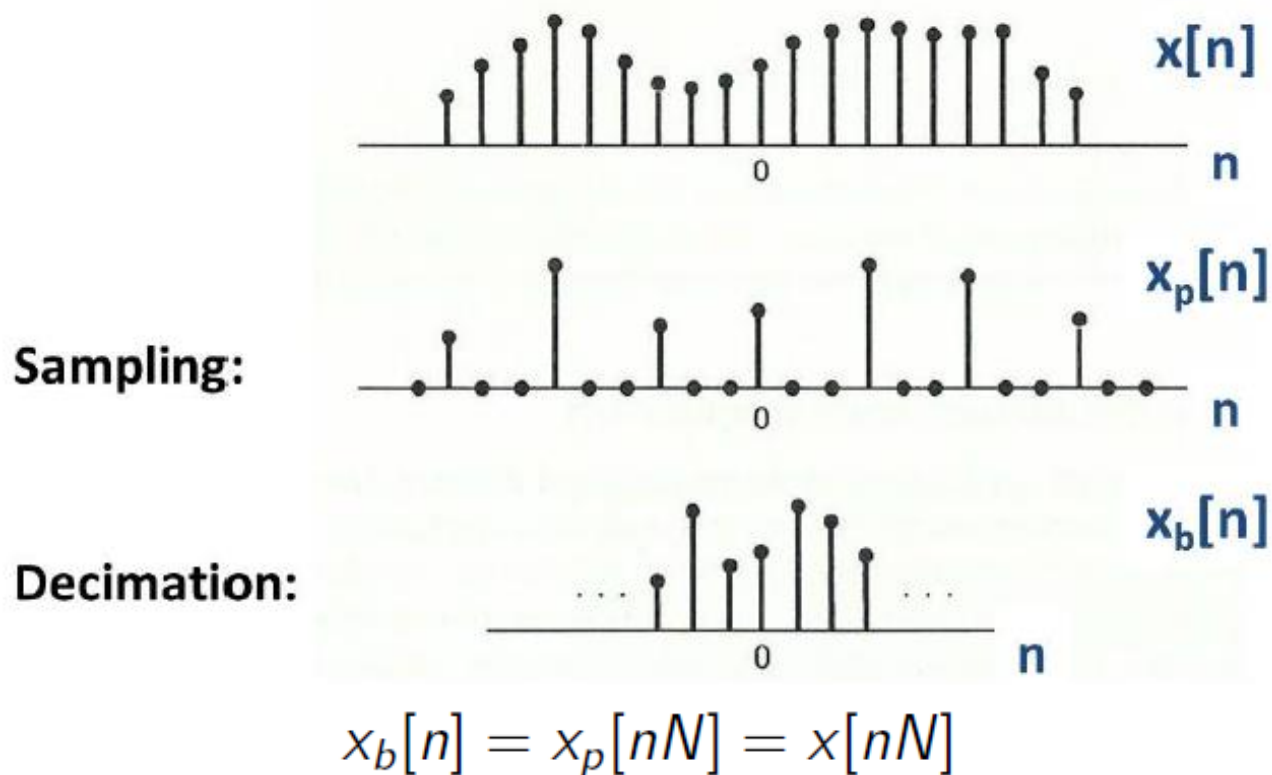
$$N, \omega_s = 2\pi/N$$

$$\omega_s > 2\omega_M$$

$$[-\omega_s/2 \quad \omega_s/2], \text{ gain } N$$

Decimation

- After sampling, there will be a great amount of redundancy
- **Decimation**: discrete-time sampling + remove zeros

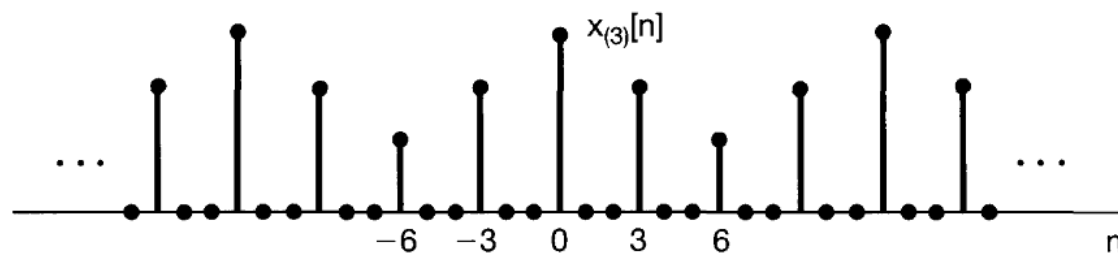
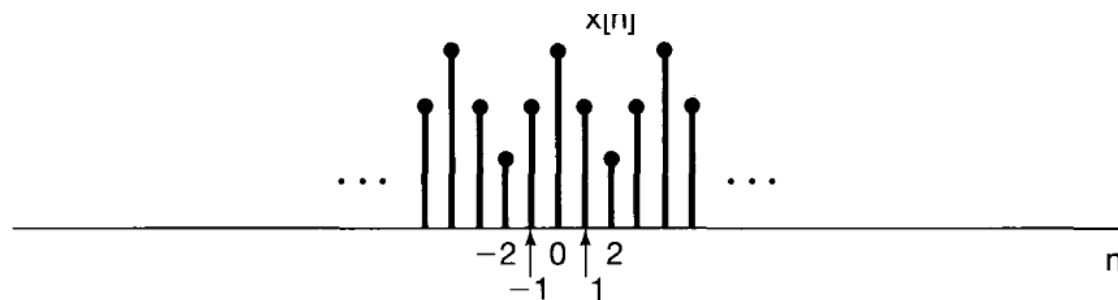


What is the relation between $X(e^{j\omega})$ and $X_b(e^{j\omega})$?

- Time Expansion

► Define $x_{(k)}[n] = \begin{cases} x[n/k], & \text{if } n \text{ is a multiple of } k \\ 0, & \text{Otherwise} \end{cases}$, then

$$x_{(k)}[n] \longleftrightarrow X(e^{jk\omega})$$



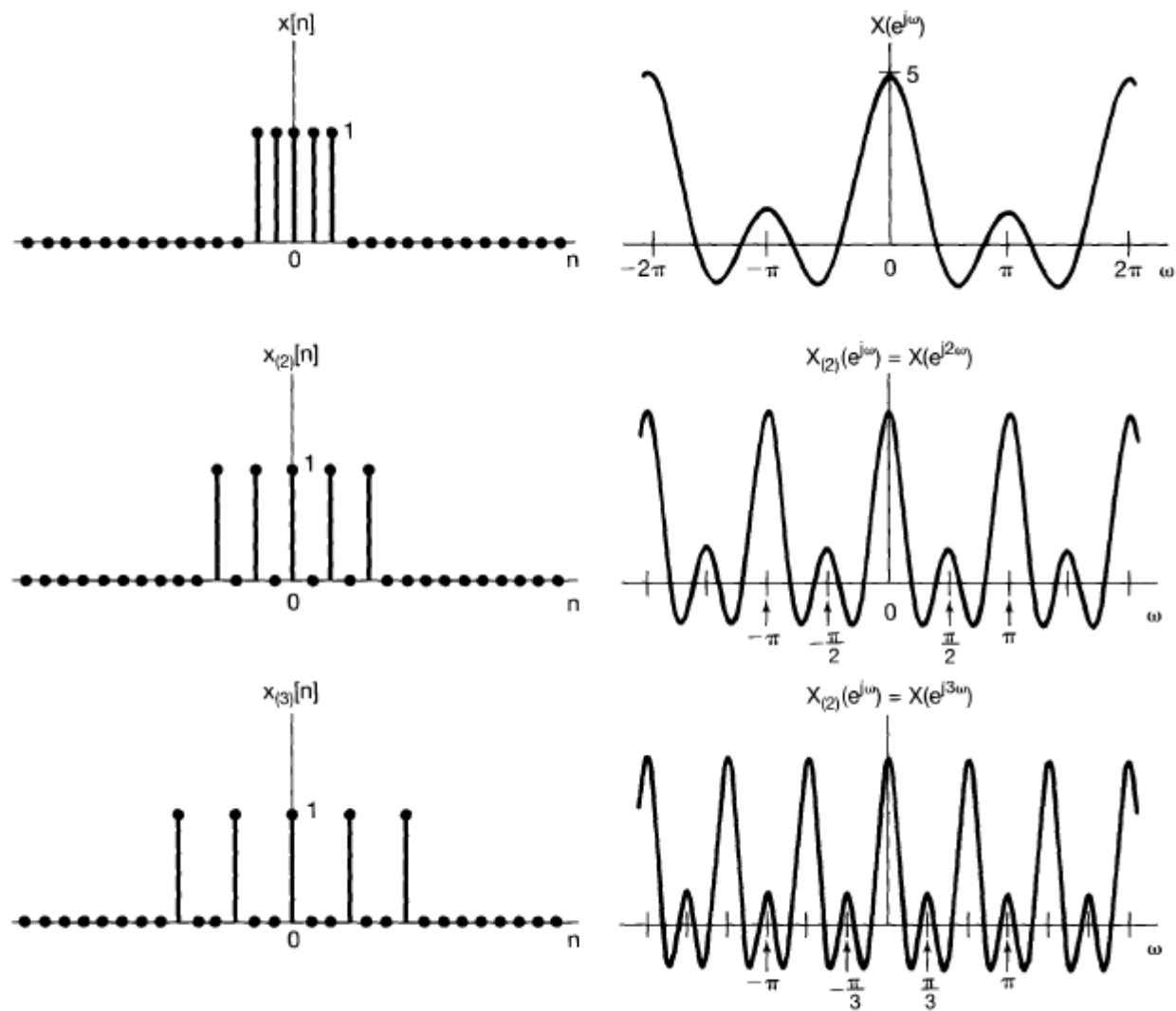


Figure 5.14 Inverse relationship between the time and frequency domains: As k increases, $x_{(k)}[n]$ spreads out while its transform is compressed.

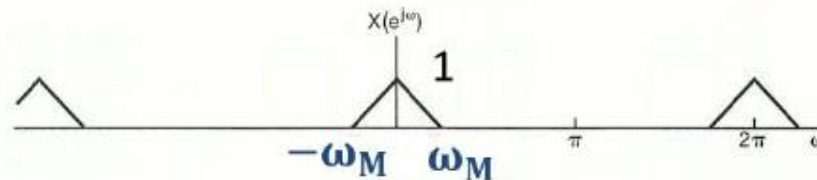
Frequency Analysis

$$\begin{aligned}
 X_b(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_b[n] e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_p[n] e^{-j\omega n/N} = X_p(e^{j\omega/N}) \\
 &= \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j\omega/N - k\omega_s})
 \end{aligned}$$

$$X_p(e^{j\omega}) = \frac{1}{N} \sum_{k=0}^{N-1} X(e^{j(\omega - k\omega_s)})$$

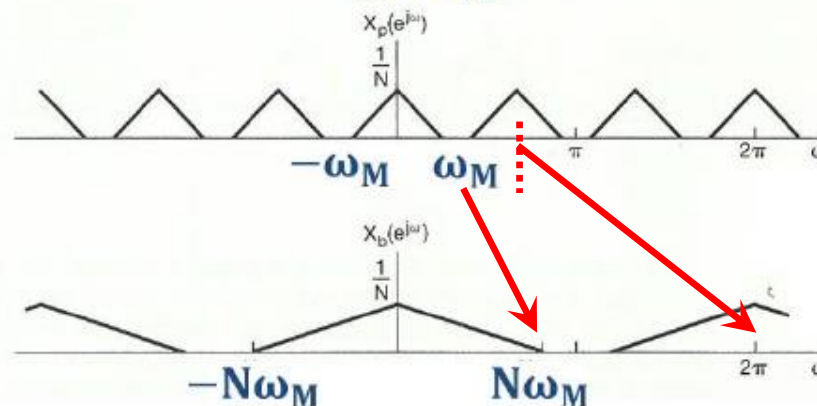
$$\omega_s = 2\pi/N$$

Sampling:



$$\omega_s = 2\pi$$

Decimation:

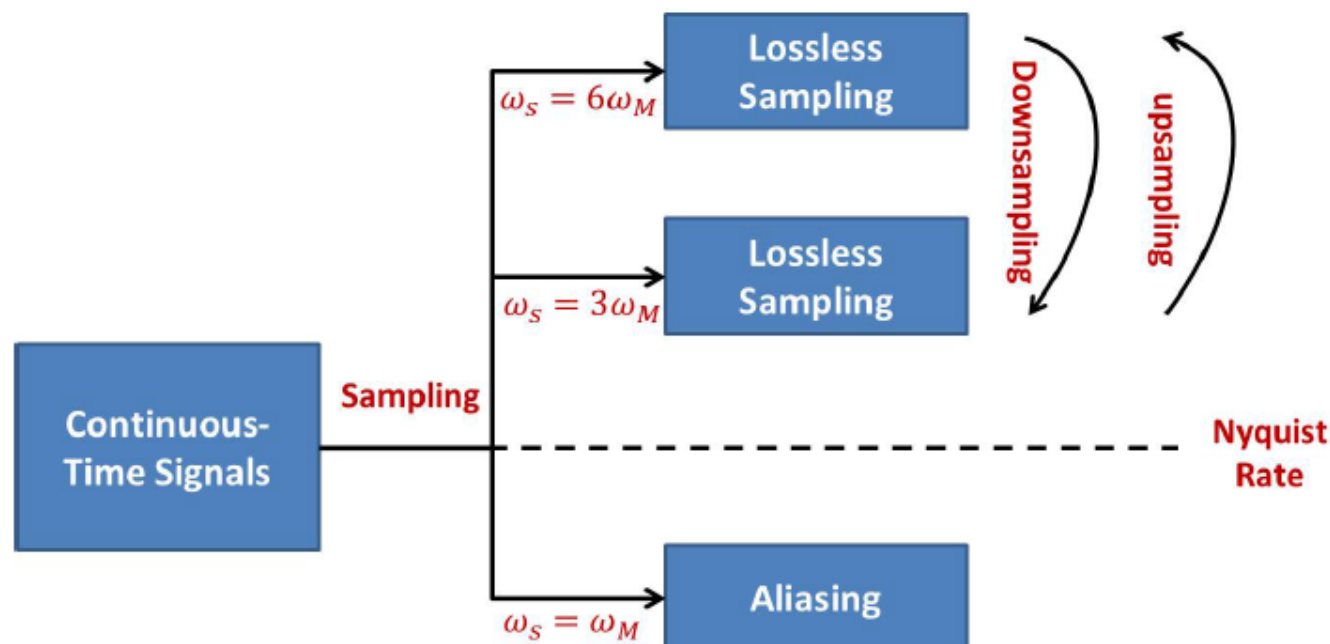


$$\omega_s = \frac{2\pi}{N}$$

$$\omega_s = 2\pi$$

Condition for Perfect Reconstruction: $N\omega_M < \pi$

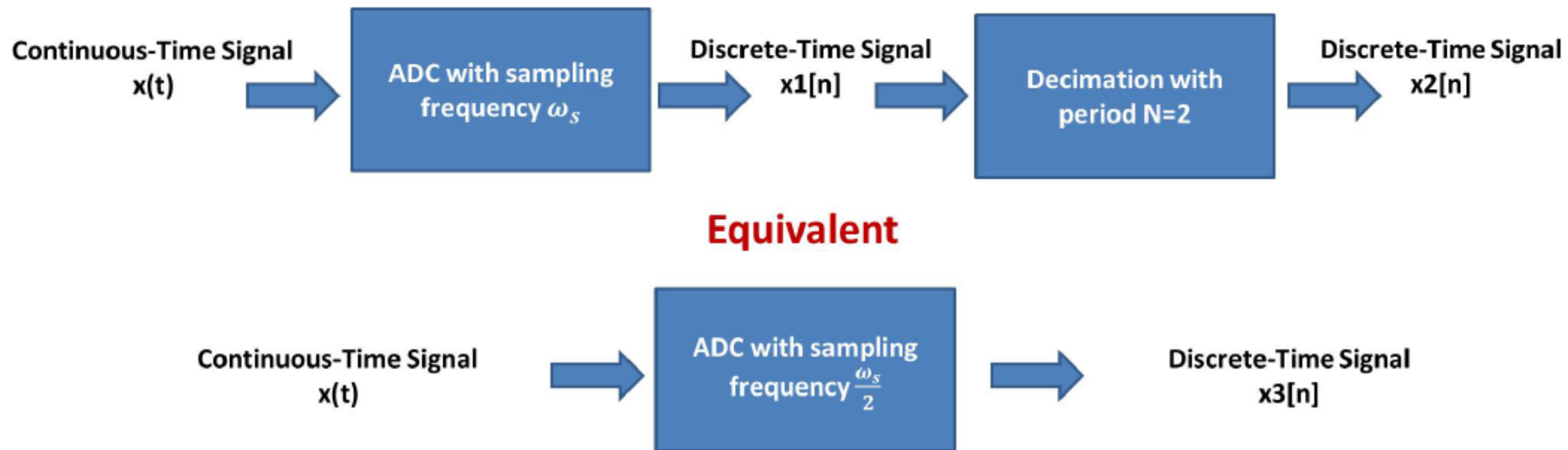
Anything Else?



- **Downsampling**: to reduce the sampling frequency (decimation)
- **Upsampling**: to generate a DT signal with higher sampling frequency
- As long as Nyquist rate is satisfied, the transform between low-sampling-frequency version and high-sampling-frequency versions is lossless.

Downsampling

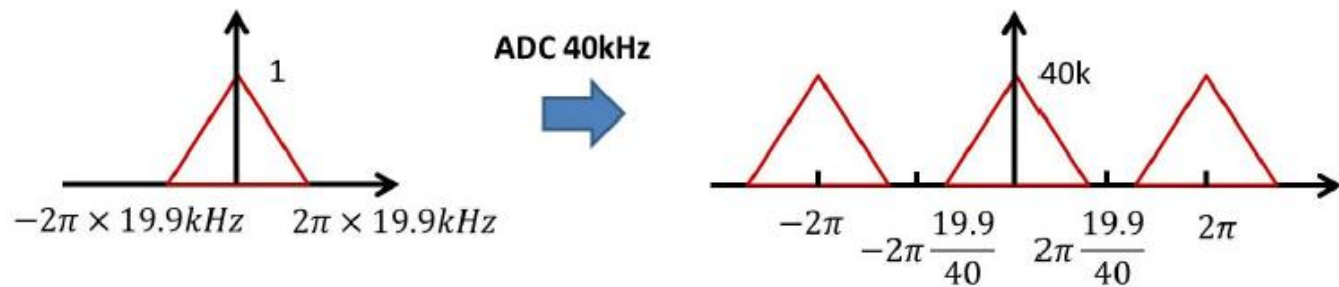
- Downsampling: a general procedure to reduce the sampling frequency



When do we need downsampling?

Downsampling Example (1/2)

- Suppose we have a clip of voice, $x(t)$, with bandwidth = 19.9kHz
- It can be converted to DT signal with sampling frequency 40kHz, denoted as $x_1[n]$



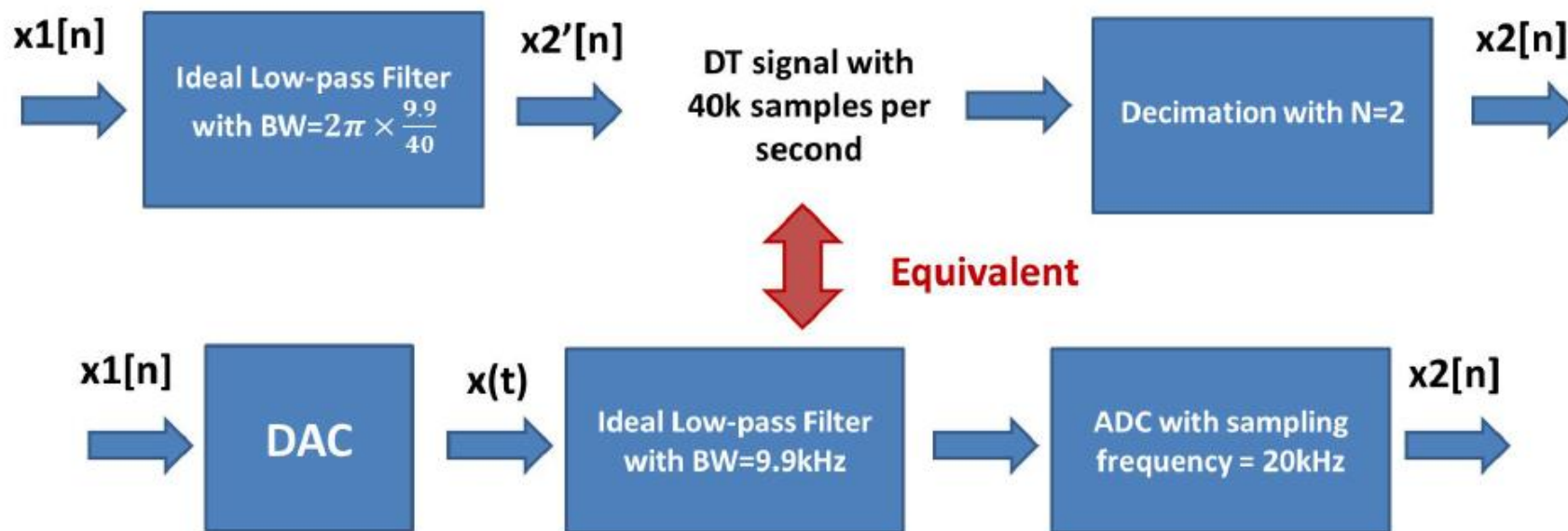
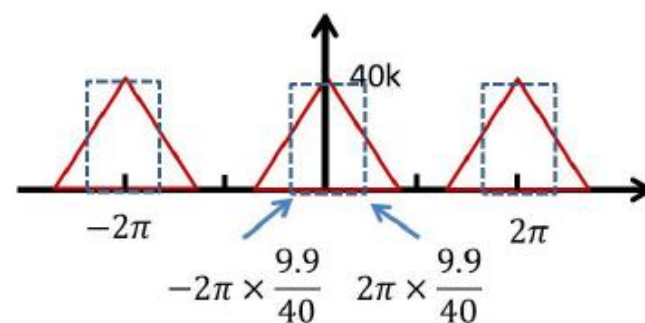
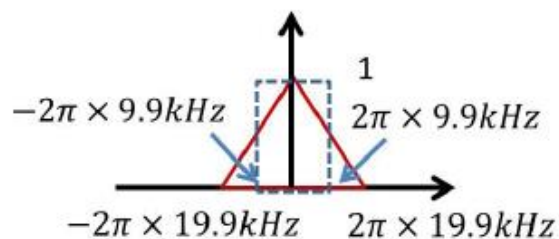
- Based on $x_1[n]$, if we want to save the voice information within 9.9kHz into another DT signal, what can we do?

One Choice:



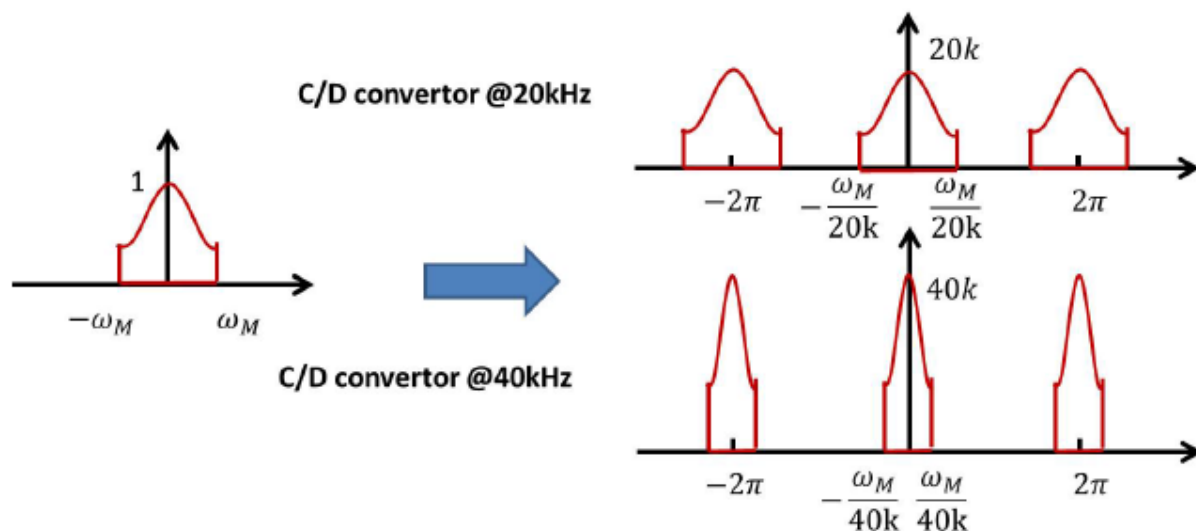
How can we generate $x_2[n]$ in discrete-time domain?

Downsampling Example (2/2)



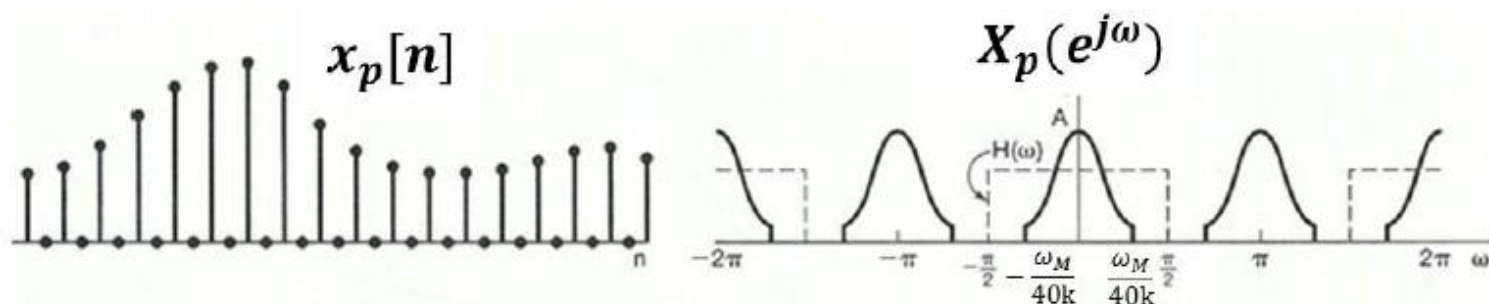
Upsampling

- **Upsampling:** a procedure to generate a sequence with higher sampling frequency
- Superpose the following two digital sound clips
 - ▶ Audio clip 1: Bandwidth= 19.9kHz, sampled at 40kHz
 - ▶ Audio clip 2: Bandwidth= 9.9kHz, sampled at 20kHz
- Double the sampling frequency of audio clip 2 (40kHz)
- How to do upsampling in discrete-time domain?



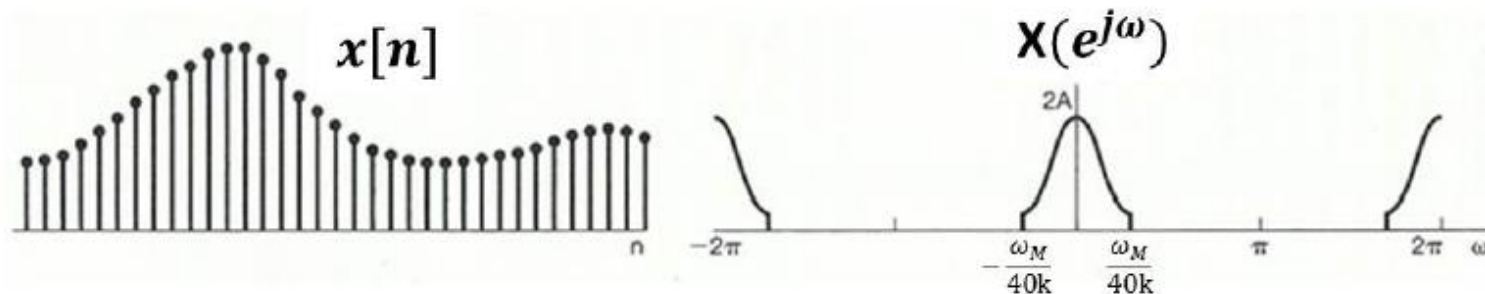
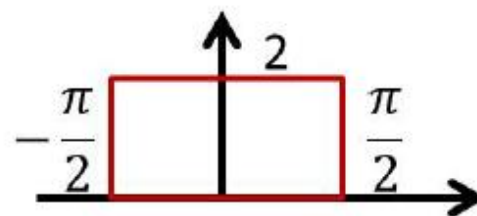
- Time expansion (Insert zeros):

$$x_p[n] = x_{b(k)}[n] \longleftrightarrow X_p(e^{j\omega}) = X_b(e^{jk\omega})$$



- Low-pass filtering:

Ideal Low-Pass Filter:

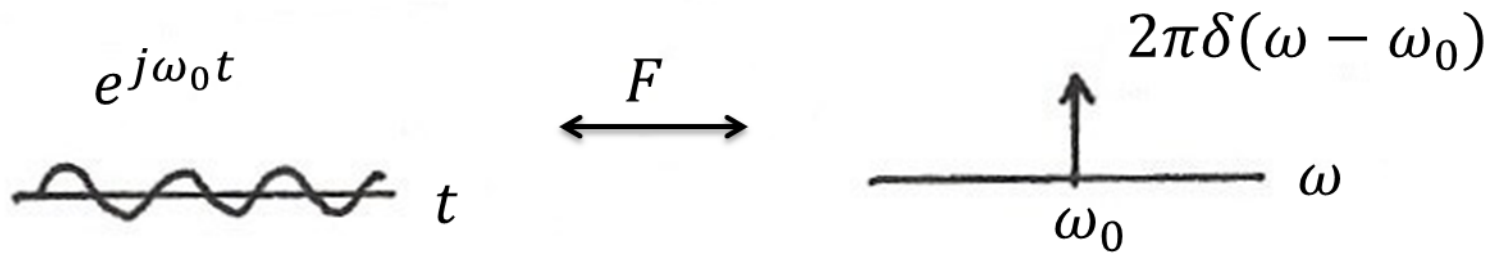




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Chapter 8

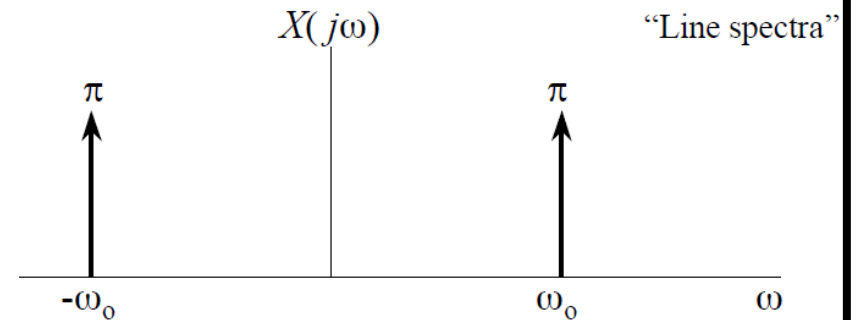
Communication Systems



$$x(t) = \cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t}$$

$\Updownarrow$

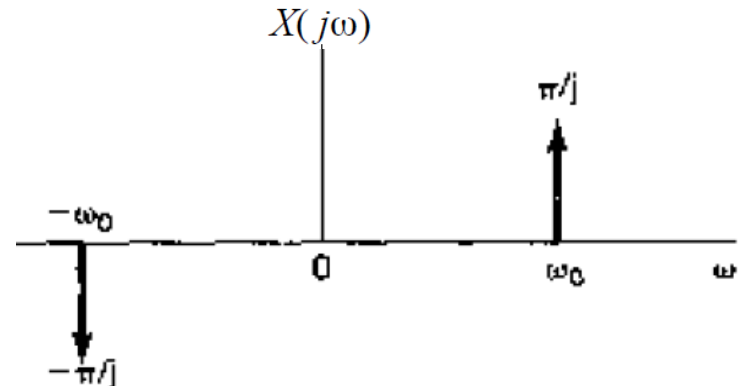
$$X(j\omega) = \pi\delta(\omega - \omega_0) + \pi\delta(\omega + \omega_0)$$



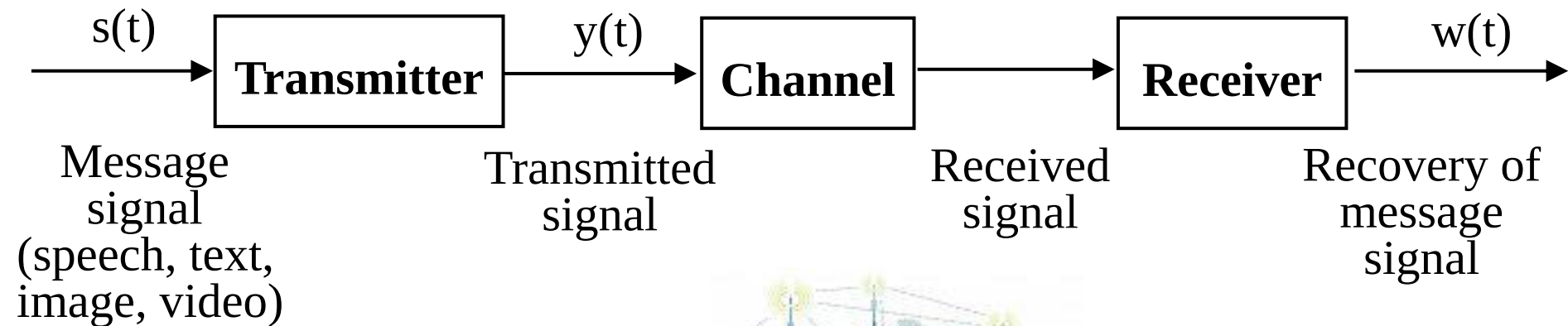
$$x(t) = \sin(\omega_0 t) = \frac{1}{2j} e^{j\omega_0 t} - \frac{1}{2j} e^{-j\omega_0 t}$$

$\Updownarrow$

$$X(j\omega) = \frac{\pi}{j} \delta(\omega - \omega_0) - \frac{\pi}{j} \delta(\omega + \omega_0)$$

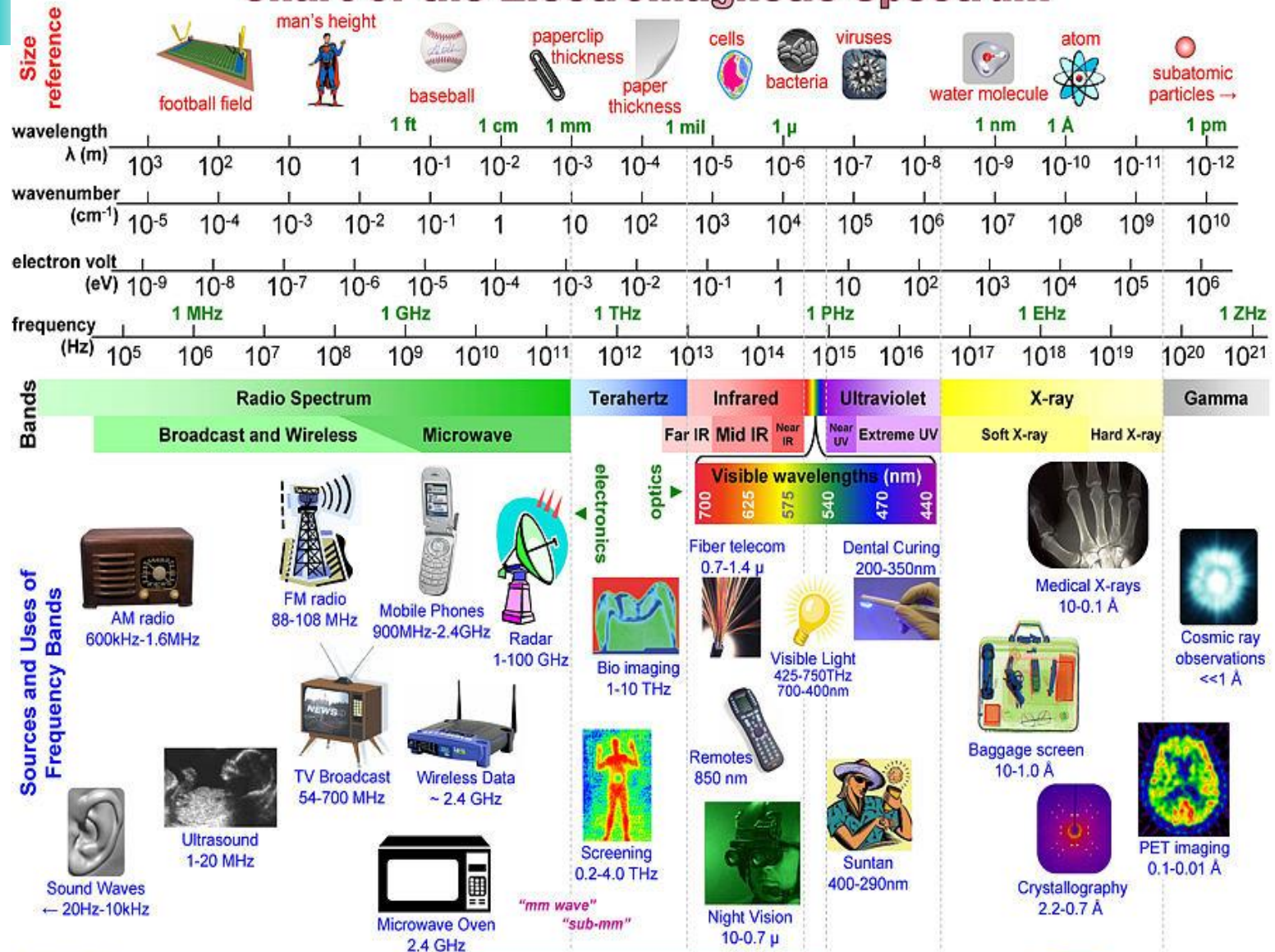


Communication Systems



Channel (media) Example

Chart of the Electromagnetic Spectrum



Amplitude modulation

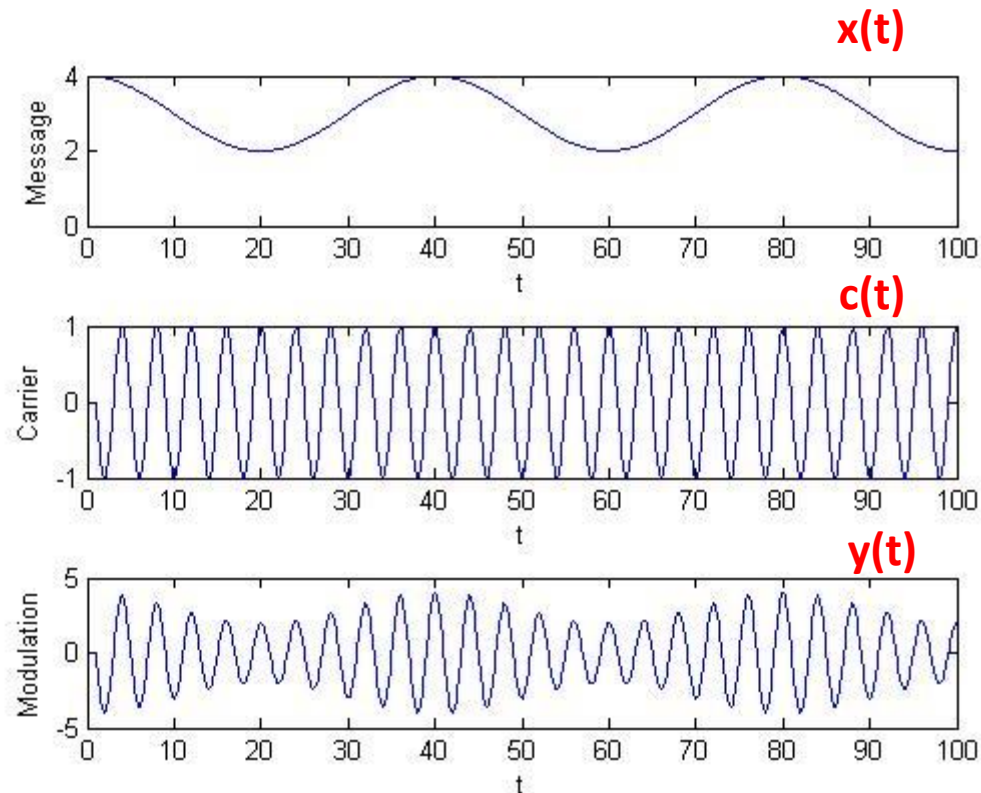
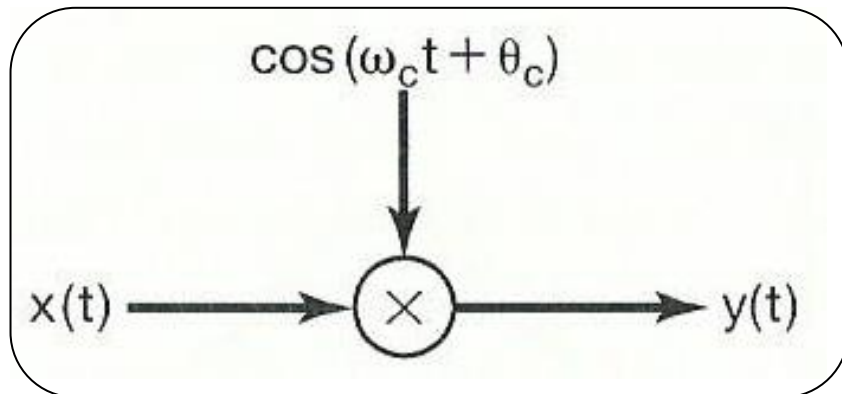
- Modulation: the general process to embed an information-bearing signal into a second signal.

$$y(t) = x(t)c(t)$$

$x(t)$: **m**odulating (or message) signal

$c(t)$: **c**arrier signal

$y(t)$: modulated signal



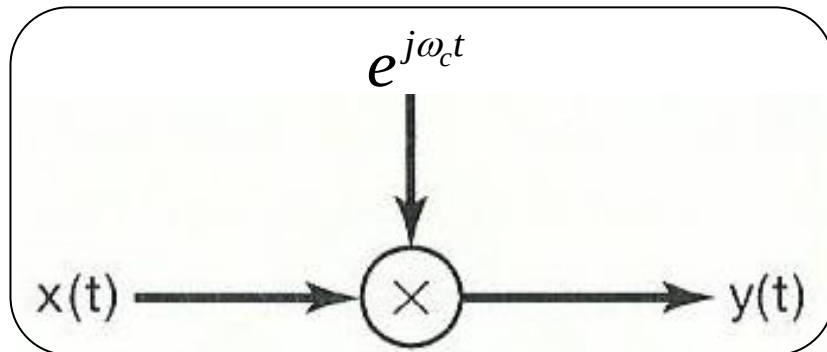
Amplitude modulation with a complex exponential carrier

$$c(t) = e^{j(\omega_c t + \theta_c)}$$

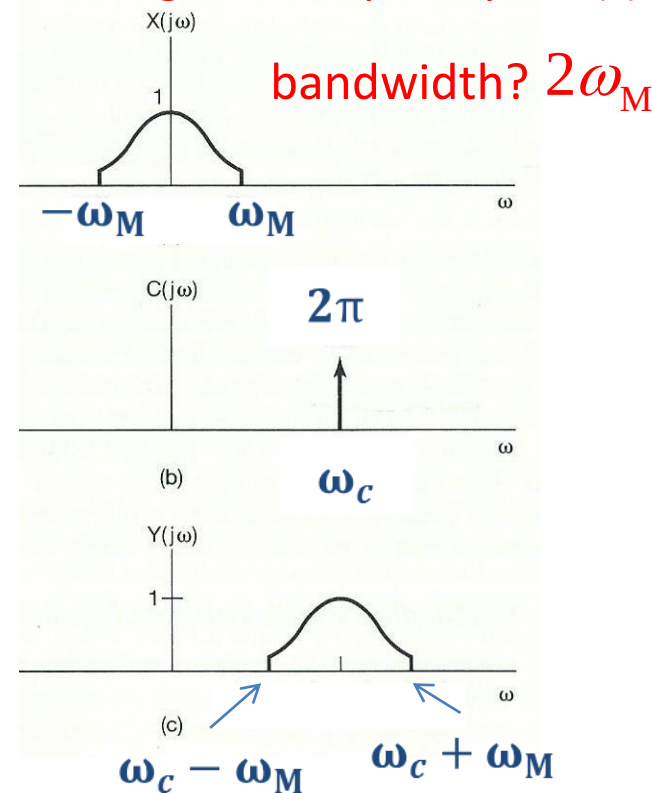
ω_c : carrier frequency

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)e^{j\omega_c t}$$



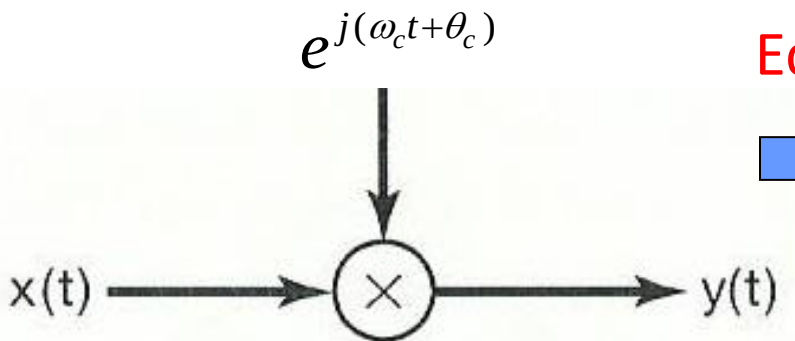
ω_M : the highest frequency in $x(t)$



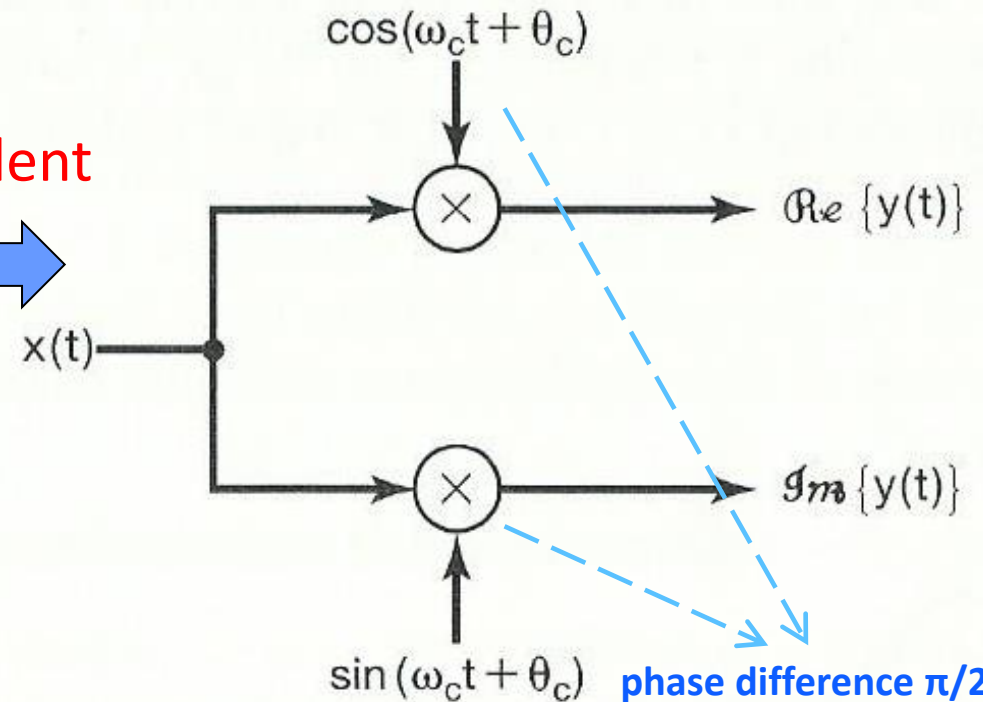
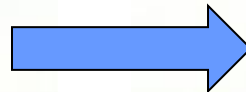
- The spectrum of the modulated output $y(t)$ is that of modulating input $x(t)$, shifted in frequency by amount of carrier frequency ω_c .
- How to recover the modulating input $x(t)$ from modulated signal $y(t)$?

Amplitude modulation with a complex exponential carrier (cont.)

$$y(t) = x(t)e^{j\omega_c t} = x(t)\cos(\omega_c t) + jx(t)\sin(\omega_c t)$$



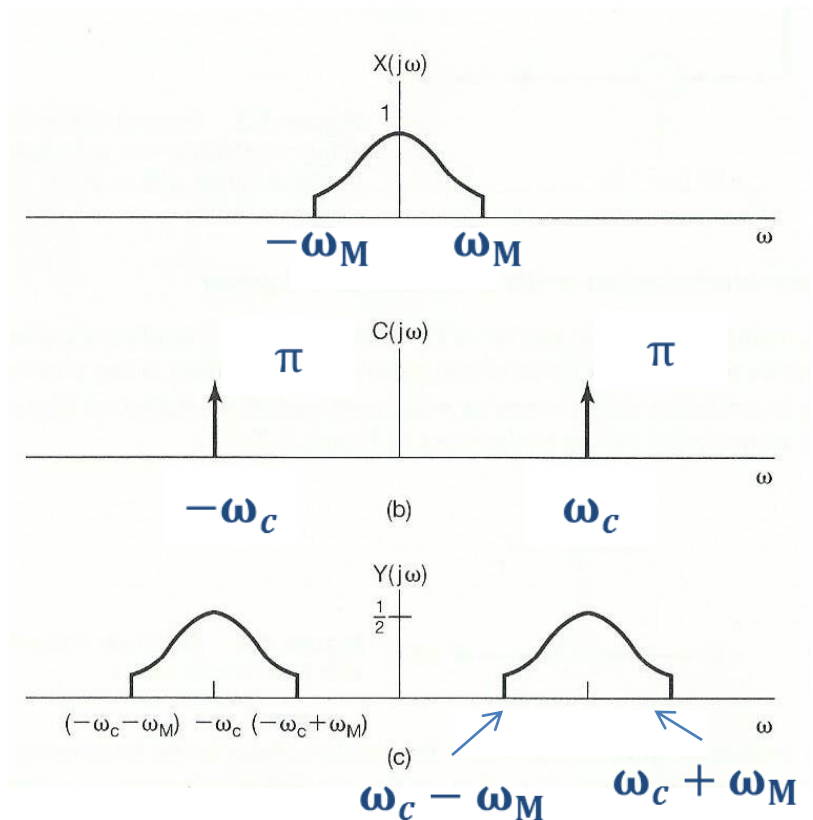
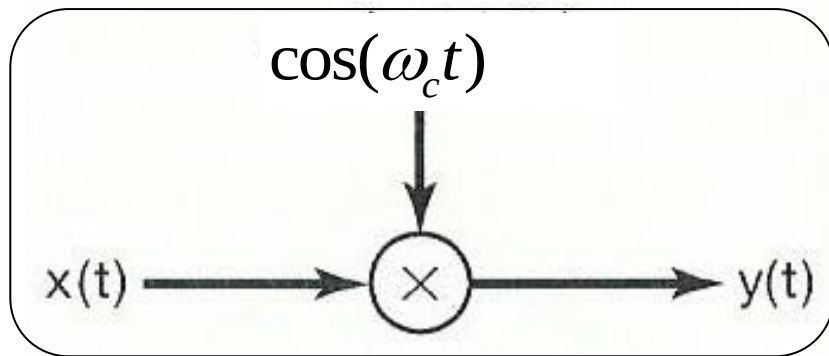
Equivalent



Amplitude modulation with a sinusoidal carrier

First, we suppose $\theta_c = 0$

$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t)$$



What is required if $x(t)$ is recoverable from $y(t)$?

Condition: $\omega_c > \omega_M$

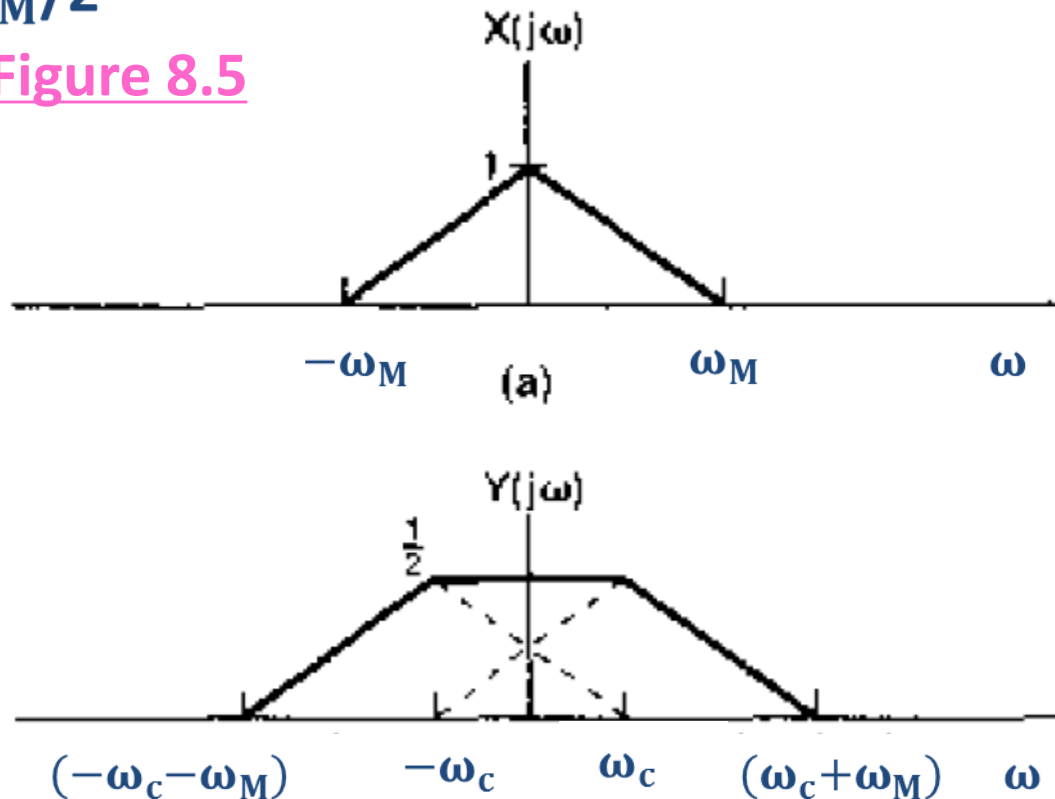
Amplitude modulation with a sinusoidal carrier (cont.)

What happens if $\omega_c < \omega_M$?

Example of an overlapping between the two replications of $X(j\omega)$

$$\omega_c = \omega_M/2$$

Figure 8.5

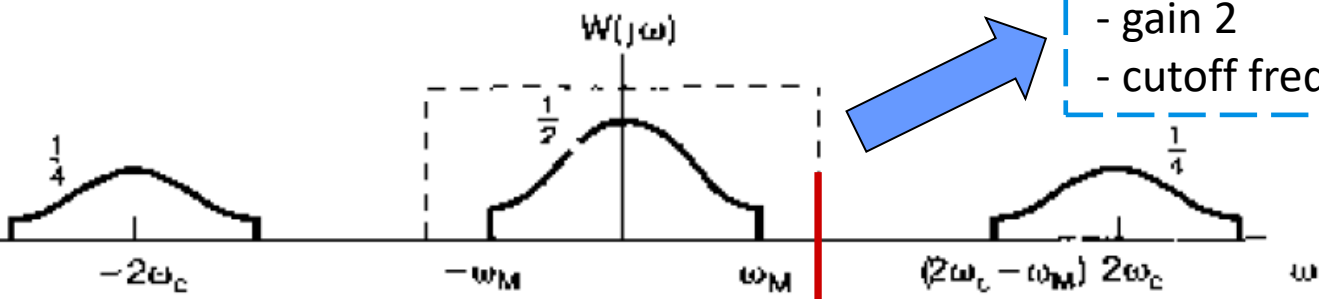
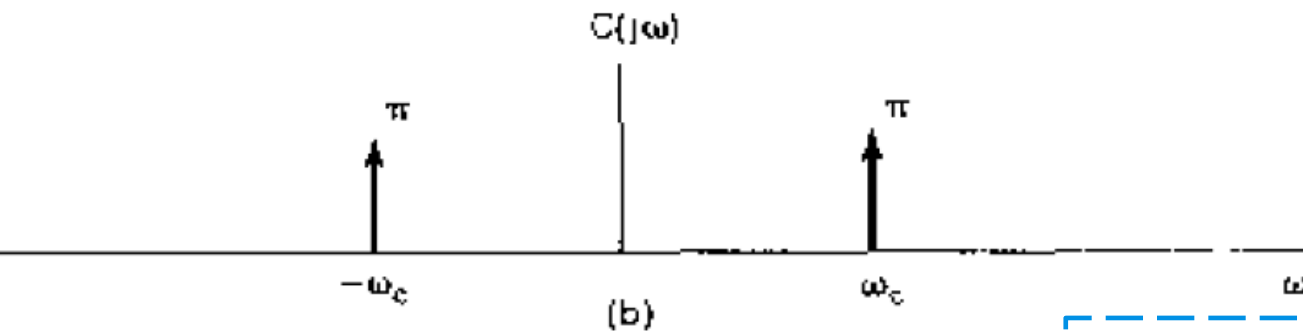
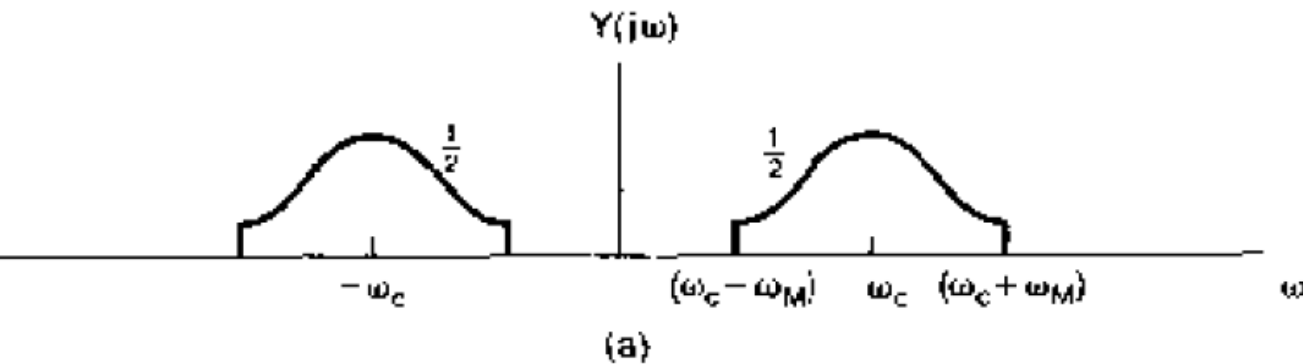


Demodulation for sinusoidal AM

- Purpose: to **recover** the information-bearing signal $x(t)$ from the modulated signal $y(t)$.
- Two types of AM demodulation:
 - ◆ Synchronous demodulation, in which transmitter and receiver are synchronized in phase and frequency
 - ◆ Asynchronous demodulation

Synchronous demodulation

With sinusoidal carrier



Ideal lowpass filtering

- gain 2

- cutoff frequency $> \omega_M$ and $< 2\omega_c - \omega_M$

$$y(t) = x(t) \cos(\omega_c t)$$



$$w(t) = y(t) \cos(\omega_c t)$$

$w(t)$: demodulated signal

Synchronous demodulation (cont.)

- Mathematically,

AM modulation $\rightarrow y(t) = x(t) \cos(\omega_c t)$

AM demodulation $\rightarrow w(t) = y(t) \cos(\omega_c t) = x(t) \cos^2(\omega_c t)$

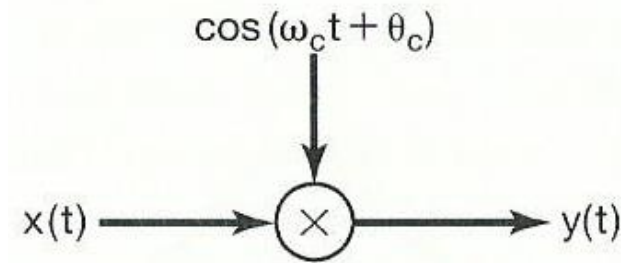
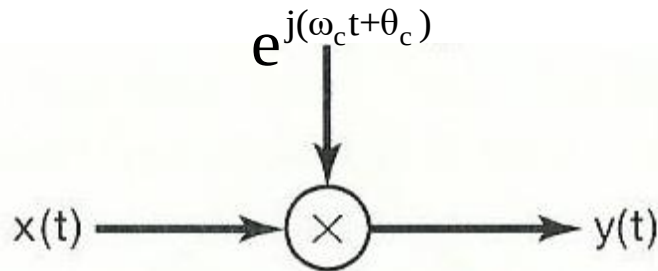
$$= x(t) \left[\frac{1}{2} + \frac{1}{2} \cos(2\omega_c t) \right]$$
$$= \boxed{\frac{1}{2} x(t)} + \frac{1}{2} x(t) \cos(2\omega_c t)$$

Synchronous demodulation (cont.)

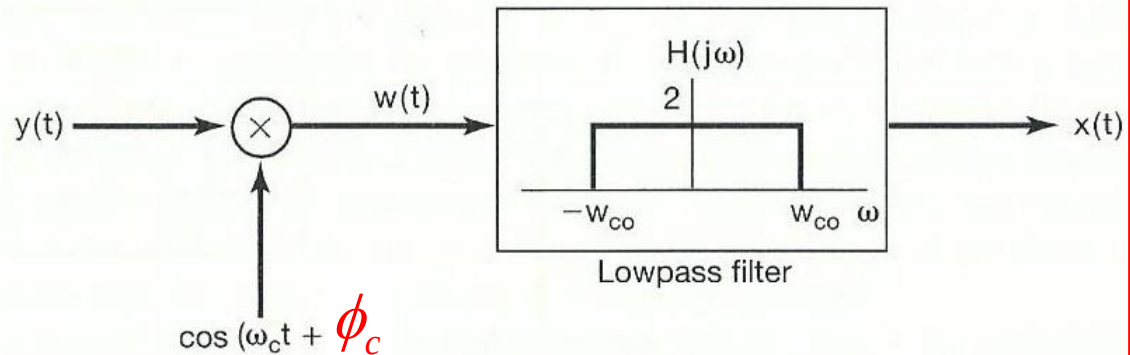
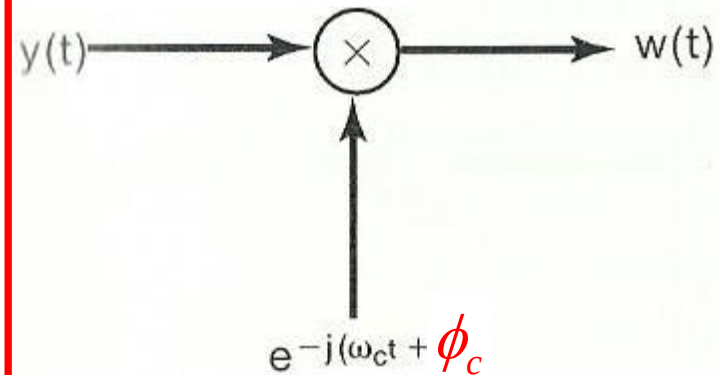
Complex exponential carrier:

Sinusoidal carrier:

AM modulation



De-modulation



$$y(t) = e^{j(\omega_c t + \theta_c)} x(t)$$

$$w(t) = e^{-j(\omega_c t + \phi_c)} y(t) \\ = e^{j(\theta_c - \phi_c)} x(t)$$

$$w(t) = x(t) \cos(\omega_c t + \theta_c) \cos(\omega_c t + \phi_c)$$

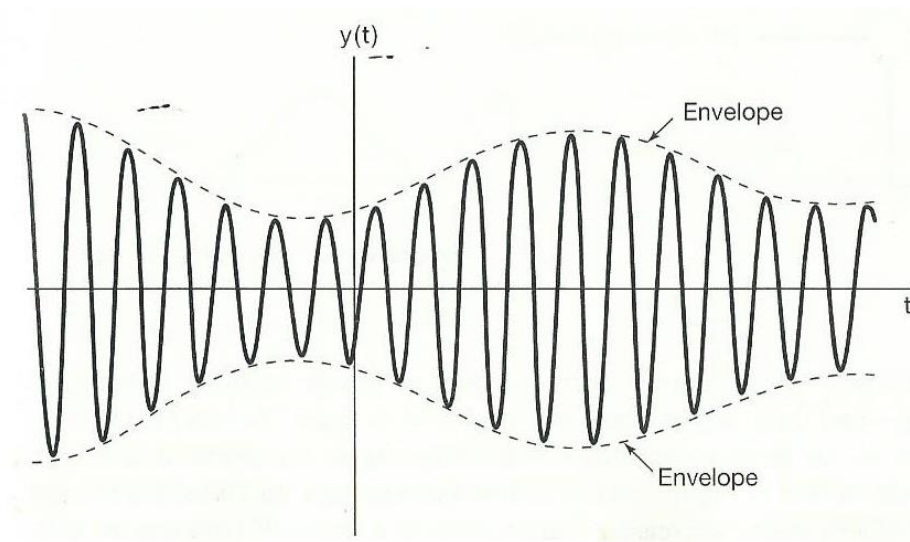
$$= \frac{1}{2} \cos(\theta_c - \phi_c) x(t) + \frac{1}{2} x(t) \cos(2\omega_c t + \theta_c + \phi_c)$$

What is the cost?

Asynchronous demodulation

- Avoid the need for synchronization between modulator and demodulator.

An example of modulated signal $y(t)$, when $x(t)$ is positive, and $\omega_c > \omega_M$.

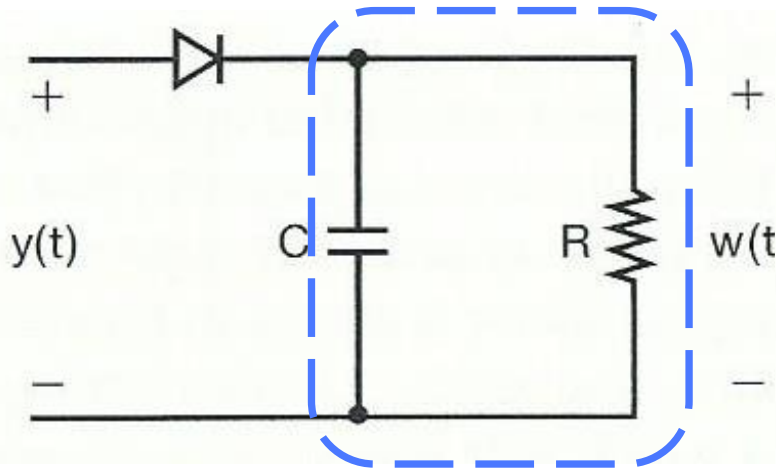


Envelope signal, a smooth curve connecting the peaks, approaches to the modulating signal $x(t)$.

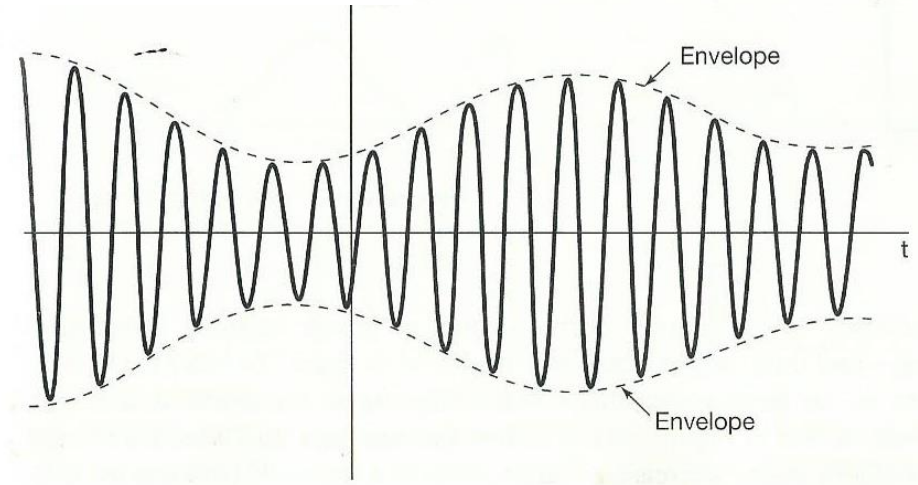
Envelope detector

Full wave or half-wave envelope detector?

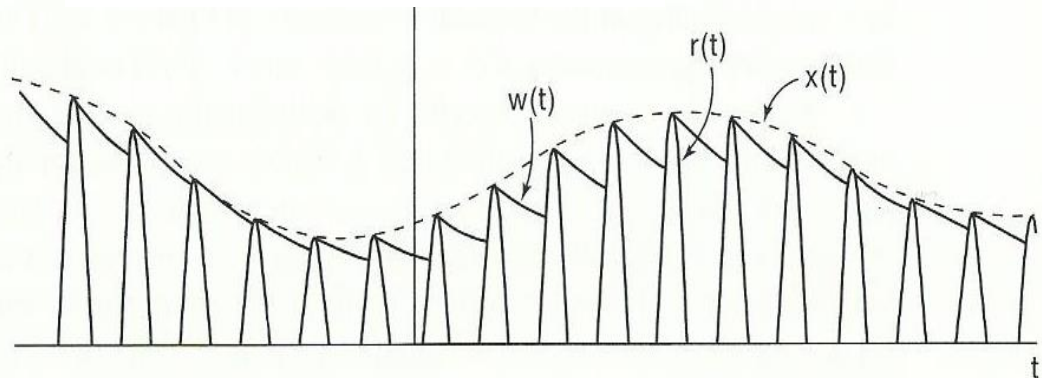
Lowpass filtering



Modulated signal

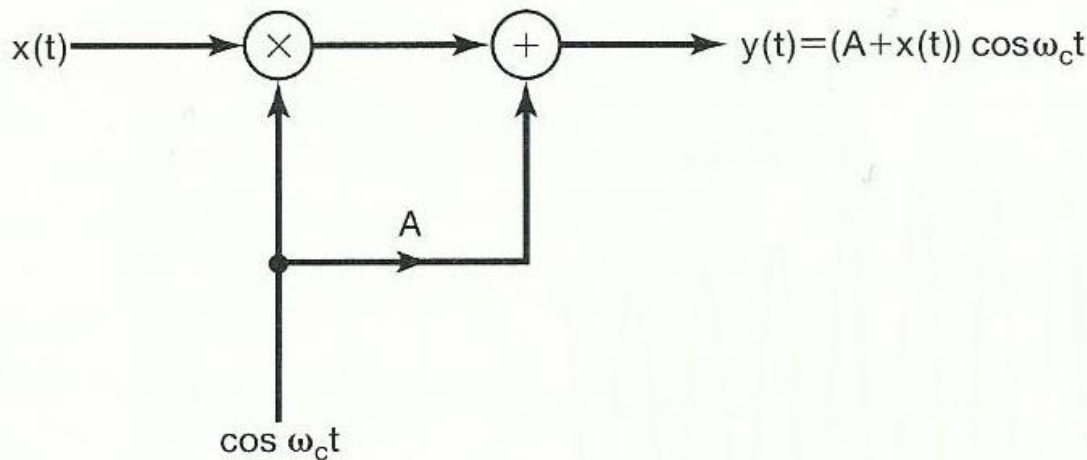


Envelope detector output

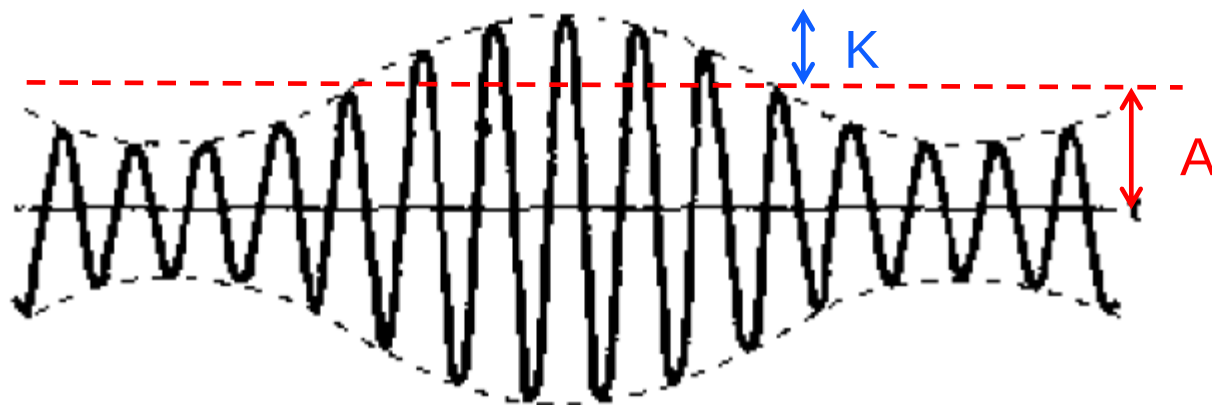


AM in an asynchronous system

- Two important assumptions:
 - ◆ ω_c is higher than ω_M
 - ◆ $x(t)$ is always positive

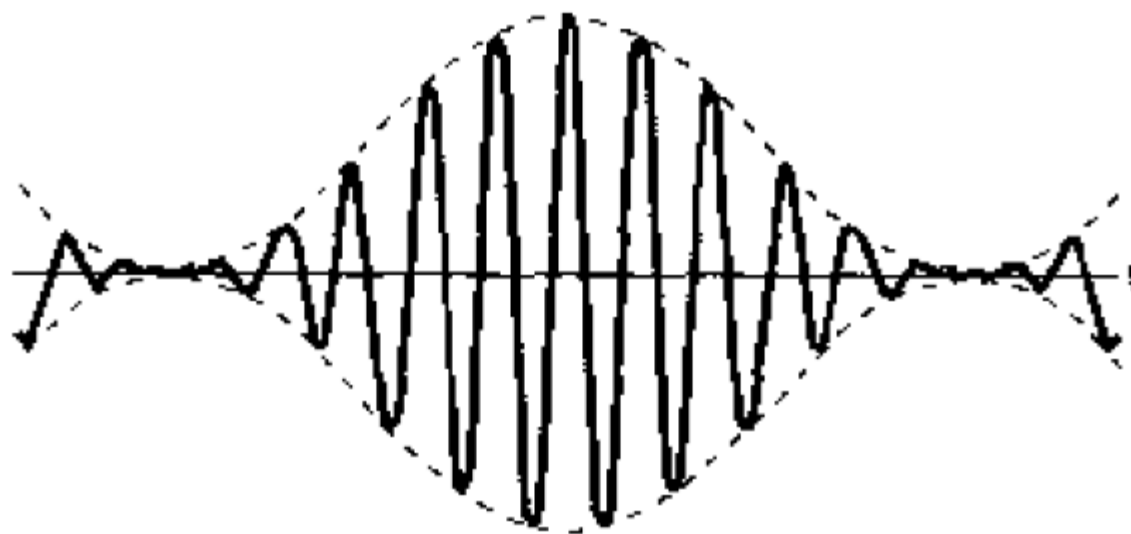


- 1) Add a constant A to make sure $[A+x(t)]$ is positive. Let K be the maximum amplitude of $x(t)$, or $|x(t)| \leq K$. We need to have $A > K$.
- 2) Define K/A as **modulation index m** , which is from 0 to 1.



$m=0.5$

(a)



$m=1$

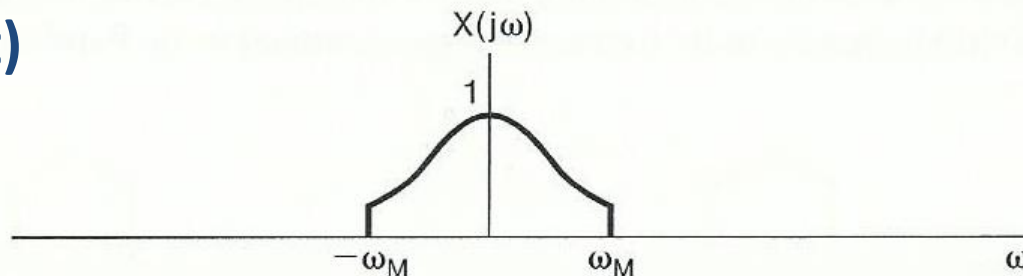
(b)

What about $m=0$?

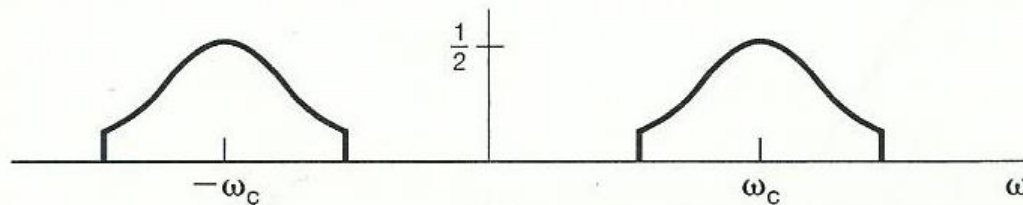
AM in an asynchronous system (cont.)

Spectrum of modulated signal in an asynchronous system

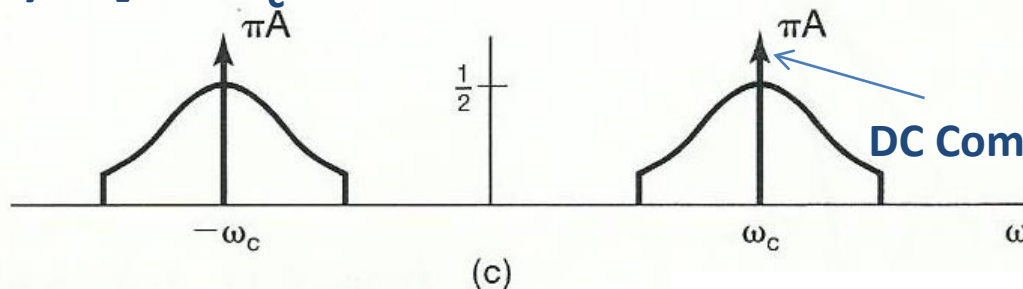
$x(t)$



$x(t)\cos\omega_c t$



$[x(t)+A]\cos\omega_c t$

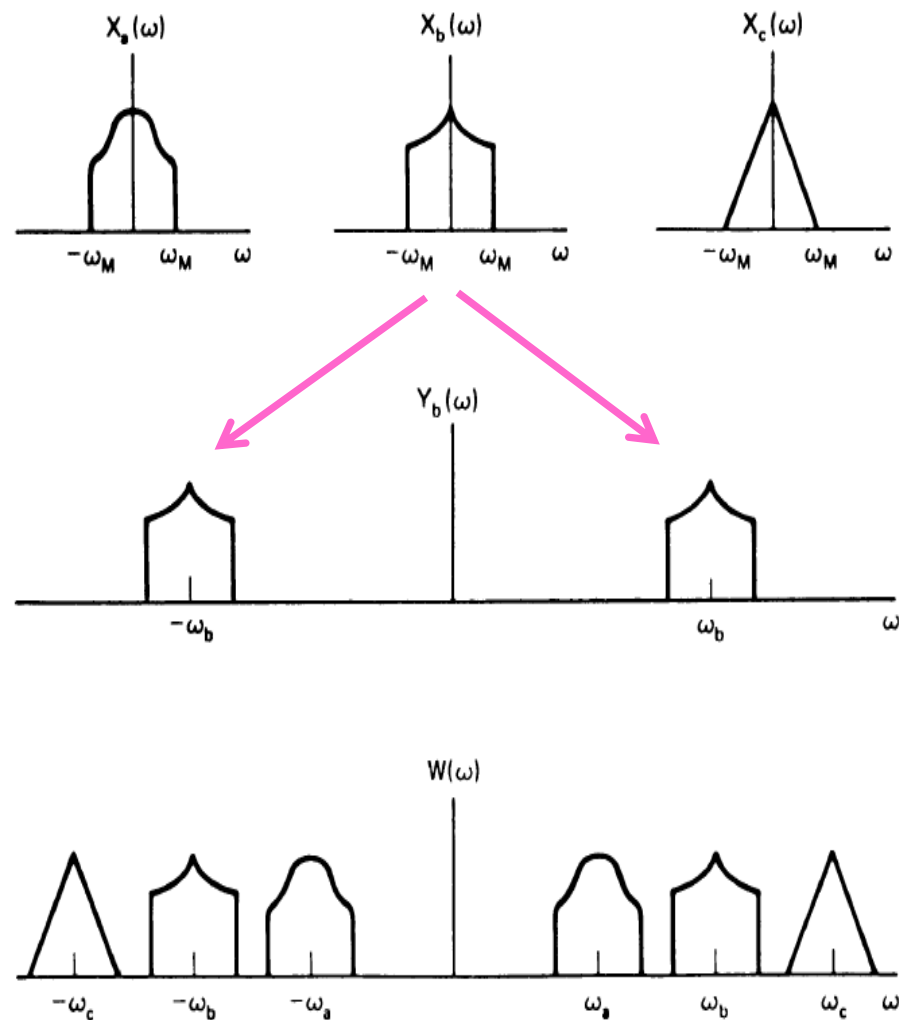
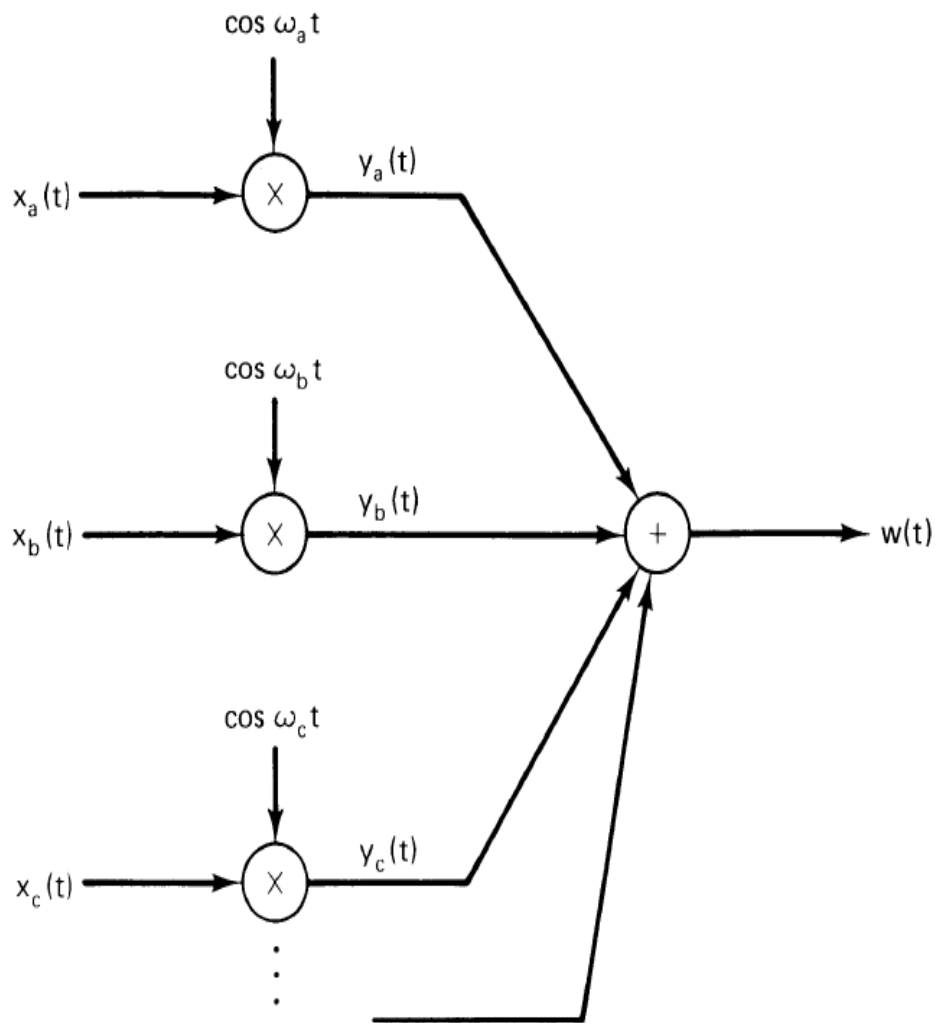


What is the cost ?

Frequency-division multiplexing (FDM)

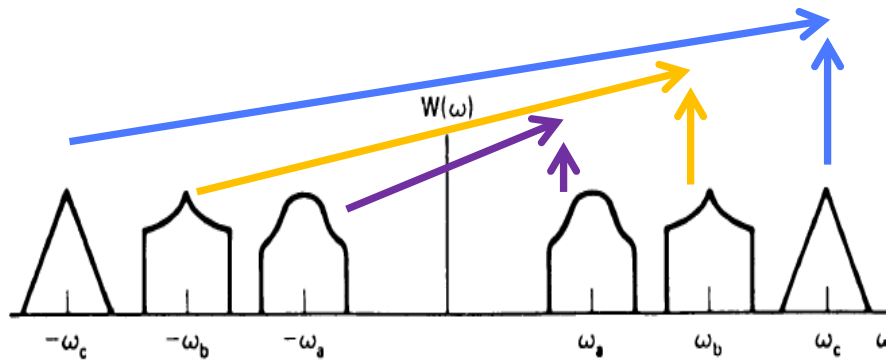
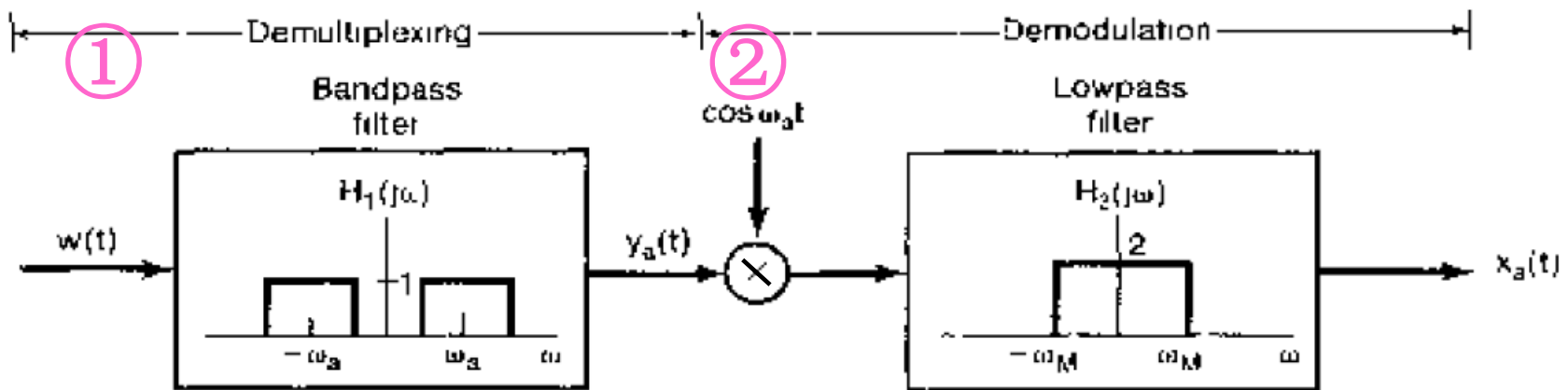
- Systems for transmitting signals provide more bandwidth than is required for one signal.
 - ◆ E.g., speech signal → 20 ~ 20 kHz
 - microwave channel → 300 MHz ~ 300 GHz
 - satellite link → a few hundred MHz ~ 40 GHz
 - (more in Fig. 8.18)
- Different modulating signals (e.g., speech), which are overlapping in frequency, can have their spectra shifted (e.g., by sinusoidal AM) without overlapping, so they can be transmitted simultaneously over a single wideband channel.

FDM (cont.)



FDM demultiplexing and demodulation

- ① bandpass filtering to have the modulated signal from one channel
- ② demodulation to recover the modulating signal

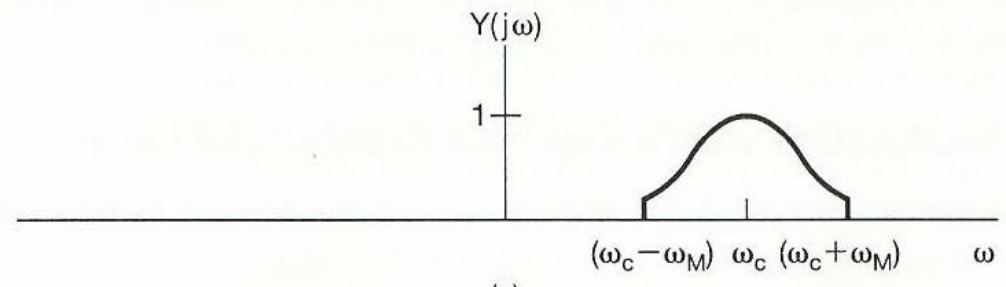
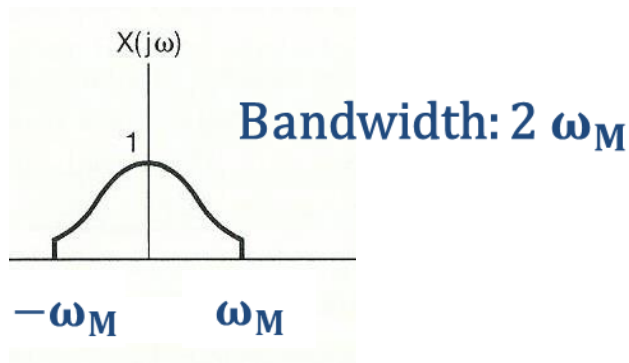


Occupy twice the original bandwidth
 → insufficient use of bandwidth

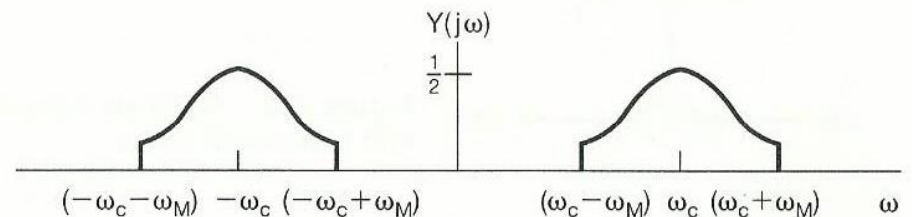
Single-sideband (SSB) sinusoidal AM

Occupied bandwidth:

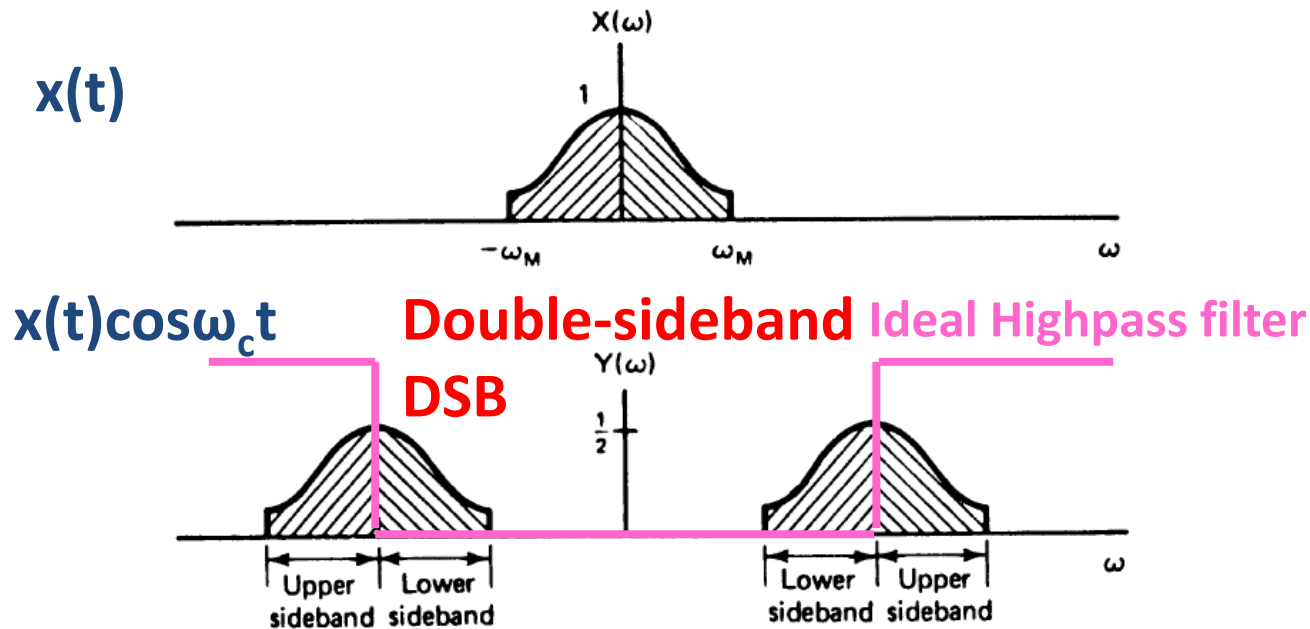
- With exponential carrier, the bandwidth is still $2\omega_M$



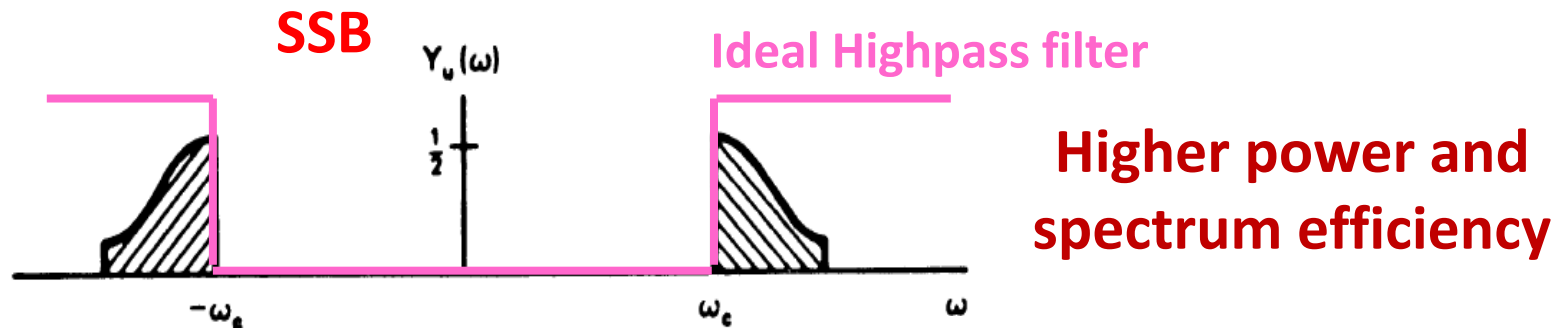
- With sinusoidal carrier, twice bandwidth is required.



SSB sinusoidal AM (cont.)

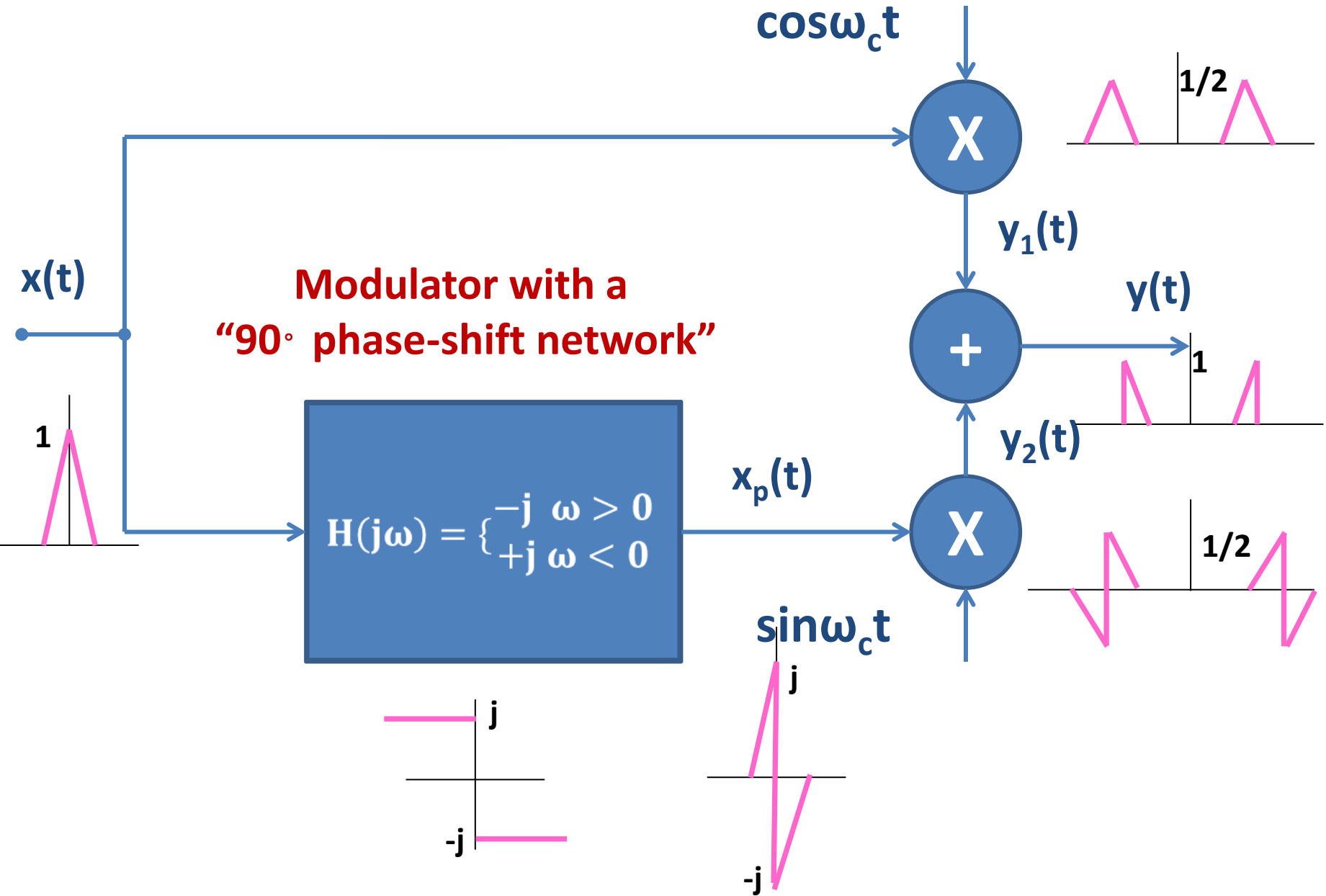


Observation: $x(t)$ can be recovered if two upper (or lower) sidebands are retained.



What is the cost ?

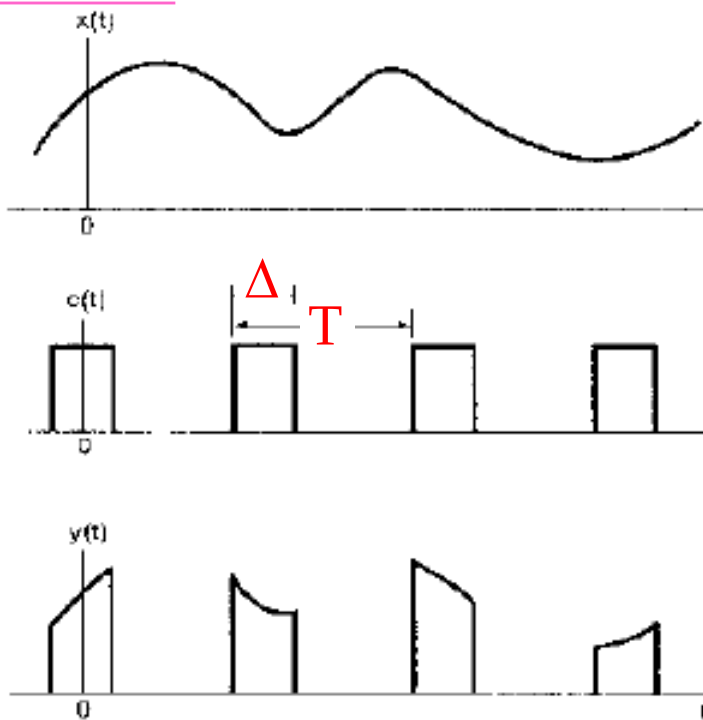
SSB sinusoidal AM (cont.)



Amplitude modulation with a pulse train

- Carrier signal could be a sinusoidal signal, or a pulse train.

Figure 8.23



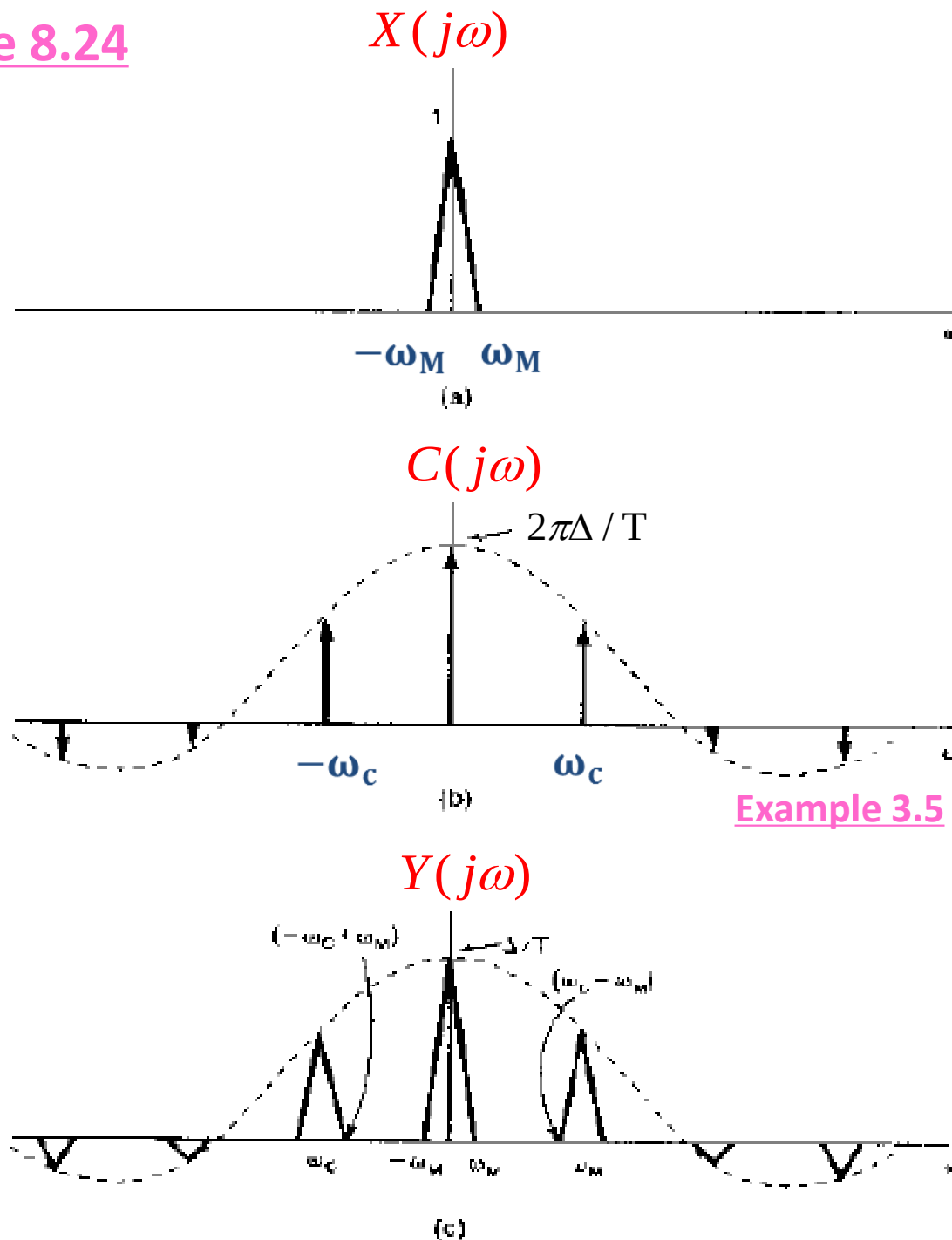
$$X(j\omega)$$

$$C(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k \delta(\omega - k\omega_c)$$

$$a_k = \frac{\sin(k\omega_c \Delta / 2)}{\pi k}$$

[Read Example 3.5](#)

$$Y(j\omega) = \sum_{k=-\infty}^{\infty} a_k X(j(\omega - k\omega_c))$$

Figure 8.24

When $\omega_c > 2\omega_M$, $X(j\omega)$ can be recovered by lowpass filtering.

Note: similar to the condition in sampling.

Example 3.5

Summary

- Meaning of amplitude modulation
 - ◆ with a complex exponential carrier
 - ◆ with a sinusoidal carrier
- Demodulation for sinusoidal AM
 - ◆ Synchronous demodulation
 - ◆ Asynchronous demodulation, and its two important assumptions
- Frequency-division multiplexing (FDM)